

249. If $f(x)$ has three roots, then it has 1 inflection point.
250. If $f(x)$ has one inflection point, then it has three real roots.

4.6 Limits at Infinity and Asymptotes

Learning Objectives

- 4.6.1 Calculate the limit of a function as x increases or decreases without bound.
- 4.6.2 Recognize a horizontal asymptote on the graph of a function.
- 4.6.3 Estimate the end behavior of a function as x increases or decreases without bound.
- 4.6.4 Recognize an oblique asymptote on the graph of a function.
- 4.6.5 Analyze a function and its derivatives to draw its graph.

We have shown how to use the first and second derivatives of a function to describe the shape of a graph. To graph a function f defined on an unbounded domain, we also need to know the behavior of f as $x \rightarrow \pm\infty$. In this section, we define limits at infinity and show how these limits affect the graph of a function. At the end of this section, we outline a strategy for graphing an arbitrary function f .

Limits at Infinity

We begin by examining what it means for a function to have a finite limit at infinity. Then we study the idea of a function with an infinite limit at infinity. Back in [Introduction to Functions and Graphs](#), we looked at vertical asymptotes; in this section we deal with horizontal and oblique asymptotes.

Limits at Infinity and Horizontal Asymptotes

Recall that $\lim_{x \rightarrow a} f(x) = L$ means $f(x)$ becomes arbitrarily close to L as long as x is sufficiently close to a . We can extend this idea to limits at infinity. For example, consider the function $f(x) = 2 + \frac{1}{x}$. As can be seen graphically in [Figure 4.40](#) and numerically in [Table 4.2](#), as the values of x get larger, the values of $f(x)$ approach 2. We say the limit as x approaches ∞ of $f(x)$ is 2 and write $\lim_{x \rightarrow \infty} f(x) = 2$. Similarly, for $x < 0$, as the values $|x|$ get larger, the values of $f(x)$

approaches 2. We say the limit as x approaches $-\infty$ of $f(x)$ is 2 and write $\lim_{x \rightarrow -\infty} f(x) = 2$.

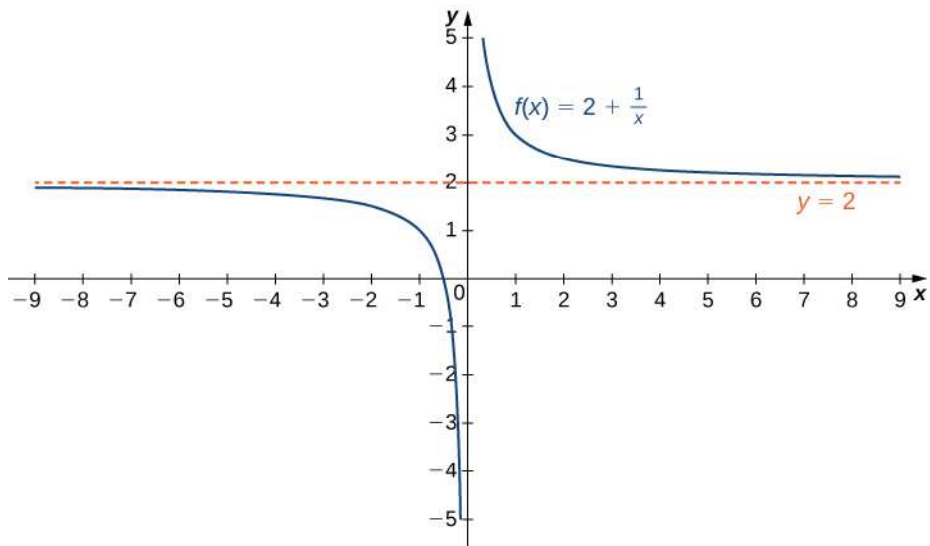


Figure 4.40 The function approaches the asymptote $y = 2$ as x approaches $\pm\infty$.

x	10	100	1,000	10,000
$2 + \frac{1}{x}$	2.1	2.01	2.001	2.0001

Table 4.2 Values of a function f as $x \rightarrow \pm\infty$

x	-10	-100	-1000	-10,000
$2 + \frac{1}{x}$	1.9	1.99	1.999	1.9999

Table 4.2 Values of a function f as $x \rightarrow \pm\infty$

More generally, for any function f , we say the limit as $x \rightarrow \infty$ of $f(x)$ is L if $f(x)$ becomes arbitrarily close to L as long as x is sufficiently large. In that case, we write $\lim_{x \rightarrow \infty} f(x) = L$. Similarly, we say the limit as $x \rightarrow -\infty$ of $f(x)$ is L if $f(x)$ becomes arbitrarily close to L as long as $x < 0$ and $|x|$ is sufficiently large. In that case, we write $\lim_{x \rightarrow -\infty} f(x) = L$. We now look at the definition of a function having a limit at infinity.

Definition

(Informal) If the values of $f(x)$ become arbitrarily close to L as x becomes sufficiently large, we say the function f has a **limit at infinity** and write

$$\lim_{x \rightarrow \infty} f(x) = L.$$

If the values of $f(x)$ become arbitrarily close to L for $x < 0$ as $|x|$ becomes sufficiently large, we say that the function f has a limit at negative infinity and write

$$\lim_{x \rightarrow -\infty} f(x) = L.$$

If the values $f(x)$ are getting arbitrarily close to some finite value L as $x \rightarrow \infty$ or $x \rightarrow -\infty$, the graph of f approaches the line $y = L$. In that case, the line $y = L$ is a horizontal asymptote of f (Figure 4.41). For example, for the function $f(x) = \frac{1}{x}$, since $\lim_{x \rightarrow \infty} f(x) = 0$, the line $y = 0$ is a horizontal asymptote of $f(x) = \frac{1}{x}$.

Definition

If $\lim_{x \rightarrow \infty} f(x) = L$ or $\lim_{x \rightarrow -\infty} f(x) = L$, we say the line $y = L$ is a **horizontal asymptote** of f .

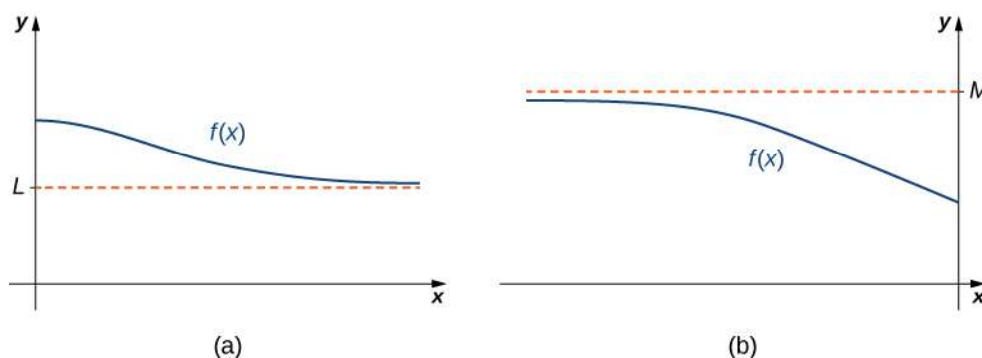


Figure 4.41 (a) As $x \rightarrow \infty$, the values of f are getting arbitrarily close to L . The line $y = L$ is a horizontal asymptote of f . (b) As $x \rightarrow -\infty$, the values of f are getting arbitrarily close to M . The line $y = M$ is a horizontal asymptote of f .

A function cannot cross a vertical asymptote because the graph must approach infinity (or $-\infty$) from at least one direction as x approaches the vertical asymptote. However, a function may cross a horizontal asymptote. In fact, a

function may cross a horizontal asymptote an unlimited number of times. For example, the function $f(x) = \frac{\cos(x)}{x} + 1$ shown in Figure 4.42 intersects the horizontal asymptote $y = 1$ an infinite number of times as it oscillates around the asymptote with ever-decreasing amplitude.

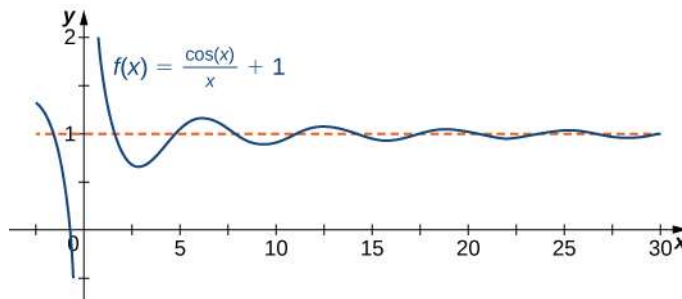


Figure 4.42 The graph of $f(x) = \frac{\cos(x)}{x} + 1$ crosses its horizontal asymptote $y = 1$ an infinite number of times.

The algebraic limit laws and squeeze theorem we introduced in [Introduction to Limits](#) also apply to limits at infinity. We illustrate how to use these laws to compute several limits at infinity.

EXAMPLE 4.21

Computing Limits at Infinity

For each of the following functions f , evaluate $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$. Determine the horizontal asymptote(s) for f .

- $f(x) = 5 - \frac{2}{x^2}$
- $f(x) = \frac{\sin x}{x}$
- $f(x) = \tan^{-1}(x)$

Solution

- Using the algebraic limit laws, we have $\lim_{x \rightarrow \infty} \left(5 - \frac{2}{x^2} \right) = \lim_{x \rightarrow \infty} 5 - 2 \left(\lim_{x \rightarrow \infty} \frac{1}{x} \right) \cdot \left(\lim_{x \rightarrow \infty} \frac{1}{x} \right) = 5 - 2 \cdot 0 = 5$.

Similarly, $\lim_{x \rightarrow -\infty} f(x) = 5$. Therefore, $f(x) = 5 - \frac{2}{x^2}$ has a horizontal asymptote of $y = 5$ and f approaches this

horizontal asymptote as $x \rightarrow \pm\infty$ as shown in the following graph.

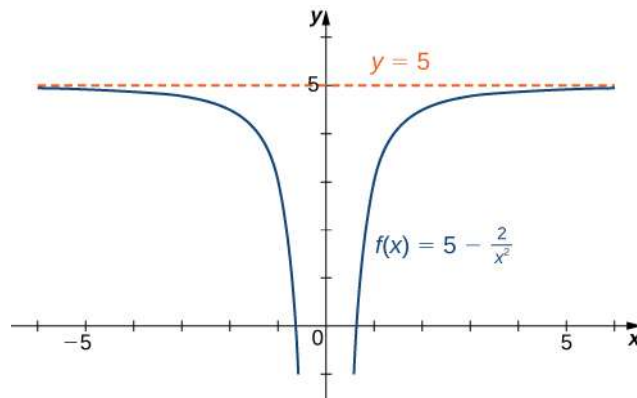


Figure 4.43 This function approaches a horizontal asymptote as $x \rightarrow \pm\infty$.

- Since $-1 \leq \sin x \leq 1$ for all x , we have

$$\frac{-1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}$$

for all $x \neq 0$. Also, since

$$\lim_{x \rightarrow \infty} \frac{-1}{x} = 0 = \lim_{x \rightarrow \infty} \frac{1}{x},$$

we can apply the squeeze theorem to conclude that

$$\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0.$$

Similarly,

$$\lim_{x \rightarrow -\infty} \frac{\sin x}{x} = 0.$$

Thus, $f(x) = \frac{\sin x}{x}$ has a horizontal asymptote of $y = 0$ and $f(x)$ approaches this horizontal asymptote as $x \rightarrow \pm\infty$ as shown in the following graph.

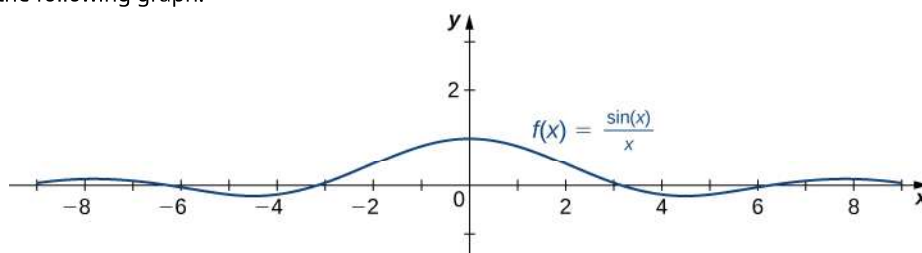


Figure 4.44 This function crosses its horizontal asymptote multiple times.

- c. To evaluate $\lim_{x \rightarrow \infty} \tan^{-1}(x)$ and $\lim_{x \rightarrow -\infty} \tan^{-1}(x)$, we first consider the graph of $y = \tan(x)$ over the interval $(-\pi/2, \pi/2)$ as shown in the following graph.

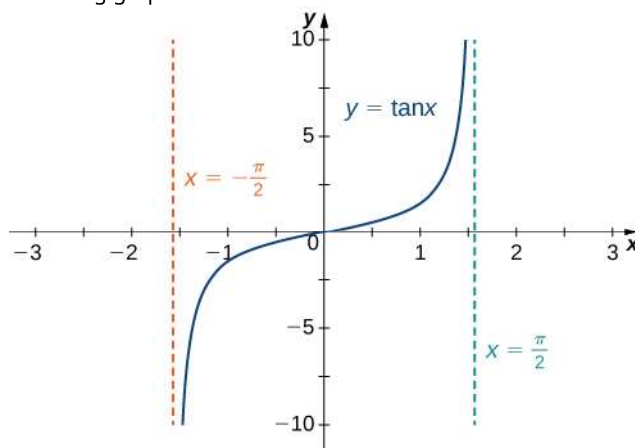


Figure 4.45 The graph of $\tan x$ has vertical asymptotes at $x = \pm\frac{\pi}{2}$

Since

$$\lim_{x \rightarrow (\pi/2)^-} \tan x = \infty,$$

it follows that

$$\lim_{x \rightarrow \infty} \tan^{-1}(x) = \frac{\pi}{2}.$$

Similarly, since

$$\lim_{x \rightarrow (-\pi/2)^+} \tan x = -\infty,$$

it follows that

$$\lim_{x \rightarrow -\infty} \tan^{-1}(x) = -\frac{\pi}{2}.$$

As a result, $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$ are horizontal asymptotes of $f(x) = \tan^{-1}(x)$ as shown in the following graph.

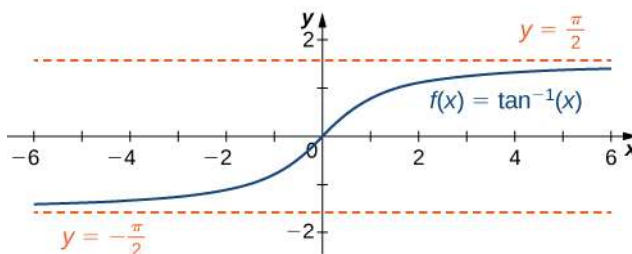


Figure 4.46 This function has two horizontal asymptotes.

✓ 4.20 Evaluate $\lim_{x \rightarrow -\infty} \left(3 + \frac{4}{x}\right)$ and $\lim_{x \rightarrow \infty} \left(3 + \frac{4}{x}\right)$. Determine the horizontal asymptotes of $f(x) = 3 + \frac{4}{x}$, if any.

Infinite Limits at Infinity

Sometimes the values of a function f become arbitrarily large as $x \rightarrow \infty$ (or as $x \rightarrow -\infty$). In this case, we write

$\lim_{x \rightarrow \infty} f(x) = \infty$ (or $\lim_{x \rightarrow -\infty} f(x) = \infty$). On the other hand, if the values of f are negative but become arbitrarily large in

magnitude as $x \rightarrow \infty$ (or as $x \rightarrow -\infty$), we write $\lim_{x \rightarrow \infty} f(x) = -\infty$ (or $\lim_{x \rightarrow -\infty} f(x) = -\infty$).

For example, consider the function $f(x) = x^3$. As seen in [Table 4.3](#) and [Figure 4.47](#), as $x \rightarrow \infty$ the values $f(x)$ become arbitrarily large. Therefore, $\lim_{x \rightarrow \infty} x^3 = \infty$. On the other hand, as $x \rightarrow -\infty$, the values of $f(x) = x^3$ are negative but

become arbitrarily large in magnitude. Consequently, $\lim_{x \rightarrow -\infty} x^3 = -\infty$.

x	10	20	50	100	1000
x^3	1000	8000	125,000	1,000,000	1,000,000,000
x	-10	-20	-50	-100	-1000
x^3	-1000	-8000	-125,000	-1,000,000	-1,000,000,000

Table 4.3 Values of a power function as $x \rightarrow \pm\infty$

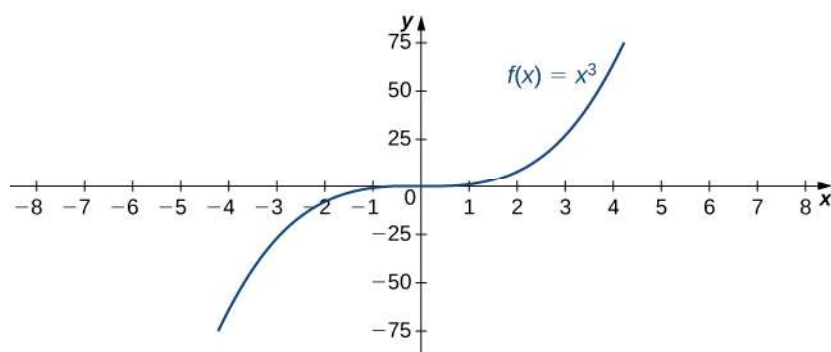


Figure 4.47 For this function, the functional values approach infinity as $x \rightarrow \pm\infty$.

Definition

(Informal) We say a function f has an infinite limit at infinity and write

$$\lim_{x \rightarrow \infty} f(x) = \infty.$$

if $f(x)$ becomes arbitrarily large for x sufficiently large. We say a function has a negative infinite limit at infinity and write

$$\lim_{x \rightarrow \infty} f(x) = -\infty.$$

if $f(x) < 0$ and $|f(x)|$ becomes arbitrarily large for x sufficiently large. Similarly, we can define infinite limits as $x \rightarrow -\infty$.

Formal Definitions

Earlier, we used the terms *arbitrarily close*, *arbitrarily large*, and *sufficiently large* to define limits at infinity informally. Although these terms provide accurate descriptions of limits at infinity, they are not precise mathematically. Here are more formal definitions of limits at infinity. We then look at how to use these definitions to prove results involving limits at infinity.

Definition

(Formal) We say a function f has a **limit at infinity**, if there exists a real number L such that for all $\varepsilon > 0$, there exists $N > 0$ such that

$$|f(x) - L| < \varepsilon$$

for all $x > N$. In that case, we write

$$\lim_{x \rightarrow \infty} f(x) = L$$

(see [Figure 4.48](#)).

We say a function f has a limit at negative infinity if there exists a real number L such that for all $\varepsilon > 0$, there exists $N < 0$ such that

$$|f(x) - L| < \varepsilon$$

for all $x < N$. In that case, we write

$$\lim_{x \rightarrow -\infty} f(x) = L.$$

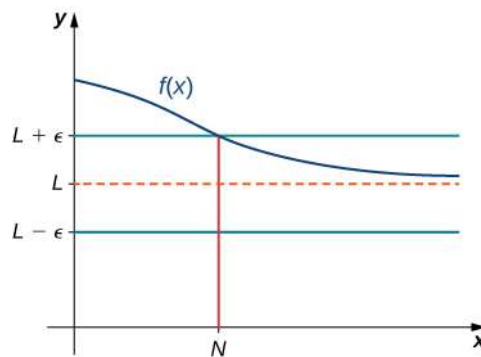


Figure 4.48 For a function with a limit at infinity, for all $x > N$, $|f(x) - L| < \epsilon$.

Earlier in this section, we used graphical evidence in [Figure 4.40](#) and numerical evidence in [Table 4.2](#) to conclude that $\lim_{x \rightarrow \infty} (2 + \frac{1}{x}) = 2$. Here we use the formal definition of limit at infinity to prove this result rigorously.

EXAMPLE 4.22

A Finite Limit at Infinity Example

Use the formal definition of limit at infinity to prove that $\lim_{x \rightarrow \infty} (2 + \frac{1}{x}) = 2$.

Solution

Let $\epsilon > 0$. Let $N = \frac{1}{\epsilon}$. Therefore, for all $x > N$, we have

$$\left| 2 + \frac{1}{x} - 2 \right| = \left| \frac{1}{x} \right| = \frac{1}{x} < \frac{1}{N} = \epsilon.$$

 4.21 Use the formal definition of limit at infinity to prove that $\lim_{x \rightarrow \infty} (3 - \frac{1}{x^2}) = 3$.

We now turn our attention to a more precise definition for an infinite limit at infinity.

Definition

(Formal) We say a function f has an **infinite limit at infinity** and write

$$\lim_{x \rightarrow \infty} f(x) = \infty$$

if for all $M > 0$, there exists an $N > 0$ such that

$$f(x) > M$$

for all $x > N$ (see [Figure 4.49](#)).

We say a function has a negative infinite limit at infinity and write

$$\lim_{x \rightarrow \infty} f(x) = -\infty$$

if for all $M < 0$, there exists an $N > 0$ such that

$$f(x) < M$$

for all $x > N$.

Similarly we can define limits as $x \rightarrow -\infty$.

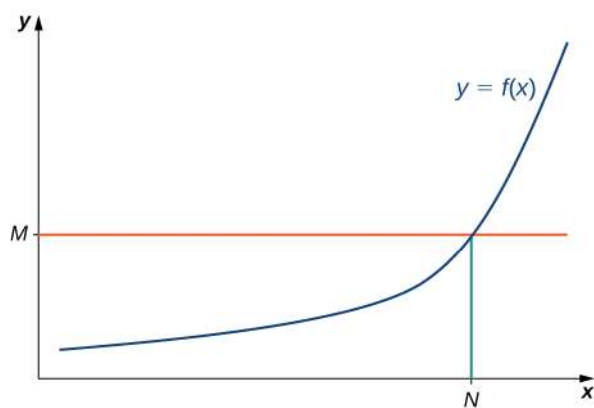


Figure 4.49 For a function with an infinite limit at infinity, for all $x > N$, $f(x) > M$.

Earlier, we used graphical evidence (Figure 4.47) and numerical evidence (Table 4.3) to conclude that $\lim_{x \rightarrow \infty} x^3 = \infty$. Here we use the formal definition of infinite limit at infinity to prove that result.

EXAMPLE 4.23

An Infinite Limit at Infinity

Use the formal definition of infinite limit at infinity to prove that $\lim_{x \rightarrow \infty} x^3 = \infty$.

✓ Solution

Let $M > 0$. Let $N = \sqrt[3]{M}$. Then, for all $x > N$, we have

$$x^3 > N^3 = (\sqrt[3]{M})^3 = M.$$

Therefore, $\lim_{x \rightarrow \infty} x^3 = \infty$.

✓ 4.22 Use the formal definition of infinite limit at infinity to prove that $\lim_{x \rightarrow \infty} 3x^2 = \infty$.

End Behavior

The behavior of a function as $x \rightarrow \pm\infty$ is called the function's **end behavior**. At each of the function's ends, the function could exhibit one of the following types of behavior:

1. The function $f(x)$ approaches a horizontal asymptote $y = L$.
2. The function $f(x) \rightarrow \infty$ or $f(x) \rightarrow -\infty$.
3. The function does not approach a finite limit, nor does it approach ∞ or $-\infty$. In this case, the function may have some oscillatory behavior.

Let's consider several classes of functions here and look at the different types of end behaviors for these functions.

End Behavior for Polynomial Functions

Consider the power function $f(x) = x^n$ where n is a positive integer. From Figure 4.50 and Figure 4.51, we see that

$$\lim_{x \rightarrow \infty} x^n = \infty; n = 1, 2, 3, \dots$$

and

$$\lim_{x \rightarrow -\infty} x^n = \begin{cases} \infty; n = 2, 4, 6, \dots \\ -\infty; n = 1, 3, 5, \dots \end{cases}.$$

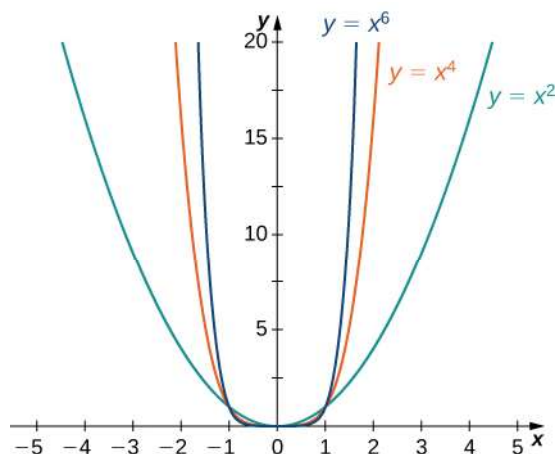


Figure 4.50 For power functions with an even exponent of n , $\lim_{x \rightarrow \infty} x^n = \infty = \lim_{x \rightarrow -\infty} x^n$.

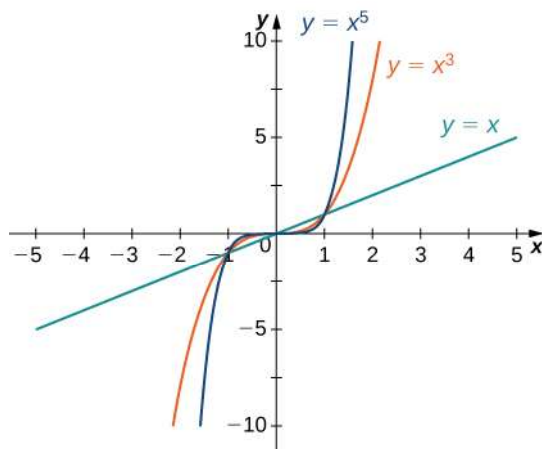


Figure 4.51 For power functions with an odd exponent of n , $\lim_{x \rightarrow \infty} x^n = \infty$ and $\lim_{x \rightarrow -\infty} x^n = -\infty$.

Using these facts, it is not difficult to evaluate $\lim_{x \rightarrow \infty} cx^n$ and $\lim_{x \rightarrow -\infty} cx^n$, where c is any constant and n is a positive integer.

If $c > 0$, the graph of $y = cx^n$ is a vertical stretch or compression of $y = x^n$, and therefore

$$\lim_{x \rightarrow \infty} cx^n = \lim_{x \rightarrow \infty} x^n \text{ and } \lim_{x \rightarrow -\infty} cx^n = \lim_{x \rightarrow -\infty} x^n \text{ if } c > 0.$$

If $c < 0$, the graph of $y = cx^n$ is a vertical stretch or compression combined with a reflection about the x -axis, and therefore

$$\lim_{x \rightarrow \infty} cx^n = -\lim_{x \rightarrow \infty} x^n \text{ and } \lim_{x \rightarrow -\infty} cx^n = -\lim_{x \rightarrow -\infty} x^n \text{ if } c < 0.$$

If $c = 0$, $y = cx^n = 0$, in which case $\lim_{x \rightarrow \infty} cx^n = 0 = \lim_{x \rightarrow -\infty} cx^n$.

EXAMPLE 4.24**Limits at Infinity for Power Functions**

For each function f , evaluate $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$.

- a. $f(x) = -5x^3$
 b. $f(x) = 2x^4$

✓ Solution

- a. Since the coefficient of x^3 is -5 , the graph of $f(x) = -5x^3$ involves a vertical stretch and reflection of the graph of $y = x^3$ about the x -axis. Therefore, $\lim_{x \rightarrow \infty} (-5x^3) = -\infty$ and $\lim_{x \rightarrow -\infty} (-5x^3) = \infty$.
- b. Since the coefficient of x^4 is 2 , the graph of $f(x) = 2x^4$ is a vertical stretch of the graph of $y = x^4$. Therefore, $\lim_{x \rightarrow \infty} 2x^4 = \infty$ and $\lim_{x \rightarrow -\infty} 2x^4 = \infty$.

✓ 4.23 Let $f(x) = -3x^4$. Find $\lim_{x \rightarrow \infty} f(x)$.

We now look at how the limits at infinity for power functions can be used to determine $\lim_{x \rightarrow \pm\infty} f(x)$ for any polynomial

function f . Consider a polynomial function

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

of degree $n \geq 1$ so that $a_n \neq 0$. Factoring, we see that

$$f(x) = a_n x^n \left(1 + \frac{a_{n-1}}{a_n} \frac{1}{x} + \dots + \frac{a_1}{a_n} \frac{1}{x^{n-1}} + \frac{a_0}{a_n} \frac{1}{x^n} \right).$$

As $x \rightarrow \pm\infty$, all the terms inside the parentheses approach zero except the first term. We conclude that

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} a_n x^n.$$

For example, the function $f(x) = 5x^3 - 3x^2 + 4$ behaves like $g(x) = 5x^3$ as $x \rightarrow \pm\infty$ as shown in [Figure 4.52](#) and [Table 4.4](#).

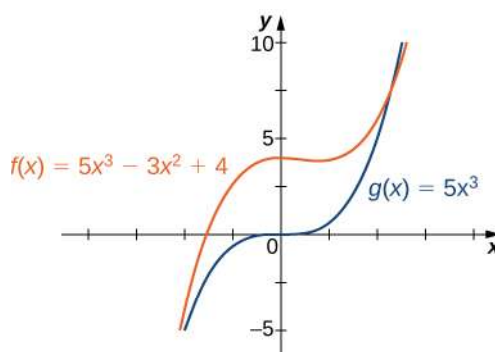


Figure 4.52 The end behavior of a polynomial is determined by the behavior of the term with the largest exponent.

x	10	100	1000
$f(x) = 5x^3 - 3x^2 + 4$	4704	4,970,004	4,997,000,004
$g(x) = 5x^3$	5000	5,000,000	5,000,000,000
x	-10	-100	-1000
$f(x) = 5x^3 - 3x^2 + 4$	-5296	-5,029,996	-5,002,999,996
$g(x) = 5x^3$	-5000	-5,000,000	-5,000,000,000

Table 4.4 A polynomial's end behavior is determined by the term with the largest exponent.

End Behavior for Algebraic Functions

The end behavior for rational functions and functions involving radicals is a little more complicated than for polynomials. In [Example 4.25](#), we show that the limits at infinity of a rational function $f(x) = \frac{p(x)}{q(x)}$ depend on the relationship between the degree of the numerator and the degree of the denominator. To evaluate the limits at infinity for a rational function, we divide the numerator and denominator by the highest power of x appearing in the denominator. This determines which term in the overall expression dominates the behavior of the function at large values of x .

EXAMPLE 4.25

Determining End Behavior for Rational Functions

For each of the following functions, determine the limits as $x \rightarrow \infty$ and $x \rightarrow -\infty$. Then, use this information to describe the end behavior of the function.

- $f(x) = \frac{3x-1}{2x+5}$ (Note: The degree of the numerator and the denominator are the same.)
- $f(x) = \frac{3x^2+2x}{4x^3-5x+7}$ (Note: The degree of numerator is less than the degree of the denominator.)
- $f(x) = \frac{3x^2+4x}{x+2}$ (Note: The degree of numerator is greater than the degree of the denominator.)

Solution

- The highest power of x in the denominator is x . Therefore, dividing the numerator and denominator by x and applying the algebraic limit laws, we see that

$$\begin{aligned}
 \lim_{x \rightarrow \pm\infty} \frac{3x-1}{2x+5} &= \lim_{x \rightarrow \pm\infty} \frac{3-1/x}{2+5/x} \\
 &= \frac{\lim_{x \rightarrow \pm\infty} (3-1/x)}{\lim_{x \rightarrow \pm\infty} (2+5/x)} \\
 &= \frac{\lim_{x \rightarrow \pm\infty} 3 - \lim_{x \rightarrow \pm\infty} 1/x}{\lim_{x \rightarrow \pm\infty} 2 + \lim_{x \rightarrow \pm\infty} 5/x} \\
 &= \frac{3-0}{2+0} = \frac{3}{2}.
 \end{aligned}$$

Since $\lim_{x \rightarrow \pm\infty} f(x) = \frac{3}{2}$, we know that $y = \frac{3}{2}$ is a horizontal asymptote for this function as shown in the following graph.

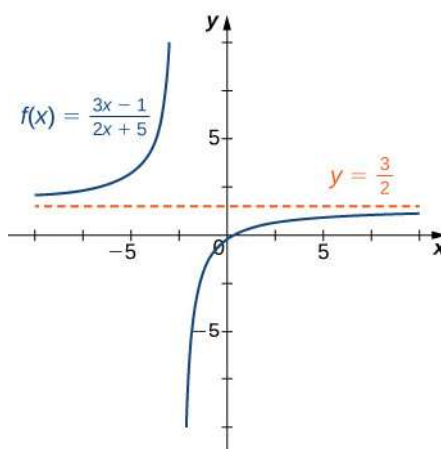


Figure 4.53 The graph of this rational function approaches a horizontal asymptote as $x \rightarrow \pm\infty$.

- b. Since the largest power of x appearing in the denominator is x^3 , divide the numerator and denominator by x^3 . After doing so and applying algebraic limit laws, we obtain

$$\lim_{x \rightarrow \pm\infty} \frac{3x^2 + 2x}{4x^3 - 5x + 7} = \lim_{x \rightarrow \pm\infty} \frac{3/x + 2/x^2}{4 - 5/x^2 + 7/x^3} = \frac{3(0) + 2(0)}{4 - 5(0) + 7(0)} = 0.$$

Therefore f has a horizontal asymptote of $y = 0$ as shown in the following graph.

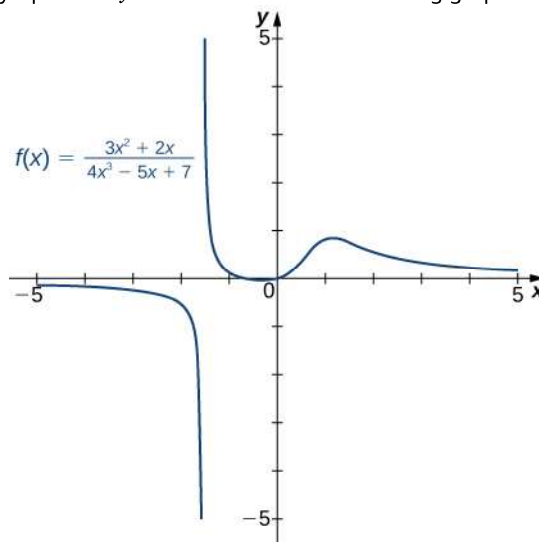


Figure 4.54 The graph of this rational function approaches the horizontal asymptote $y = 0$ as $x \rightarrow \pm\infty$.

- c. Dividing the numerator and denominator by x , we have

$$\lim_{x \rightarrow \pm\infty} \frac{3x^2 + 4x}{x + 2} = \lim_{x \rightarrow \pm\infty} \frac{3x + 4}{1 + 2/x}.$$

As $x \rightarrow \pm\infty$, the denominator approaches 1. As $x \rightarrow \infty$, the numerator approaches $+\infty$. As $x \rightarrow -\infty$, the numerator approaches $-\infty$. Therefore $\lim_{x \rightarrow \infty} f(x) = \infty$, whereas $\lim_{x \rightarrow -\infty} f(x) = -\infty$ as shown in the following figure.

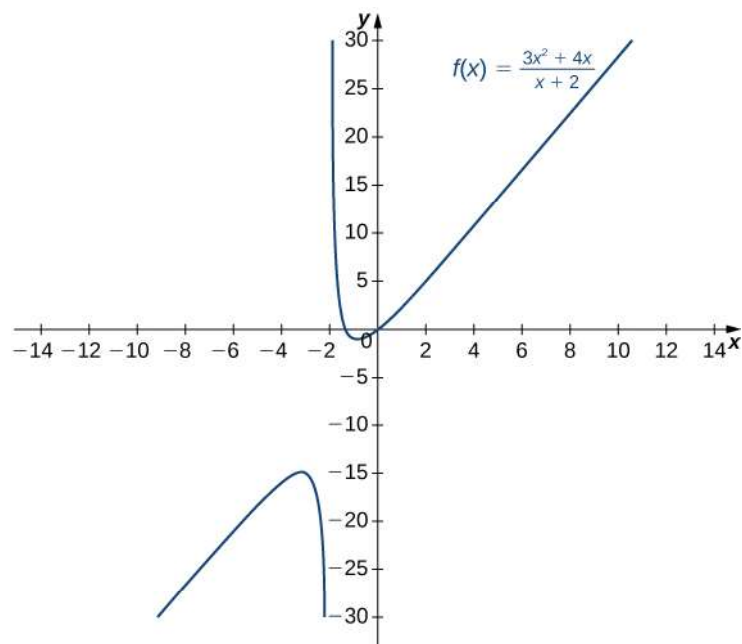


Figure 4.55 As $x \rightarrow \infty$, the values $f(x) \rightarrow \infty$. As $x \rightarrow -\infty$, the values $f(x) \rightarrow -\infty$.

- ✓ 4.24 Evaluate $\lim_{x \rightarrow \pm\infty} \frac{3x^2+2x-1}{5x^2-4x+7}$ and use these limits to determine the end behavior of $f(x) = \frac{3x^2+2x-1}{5x^2-4x+7}$.

Before proceeding, consider the graph of $f(x) = \frac{(3x^2+4x)}{(x+2)}$ shown in [Figure 4.56](#). As $x \rightarrow \infty$ and $x \rightarrow -\infty$, the graph of f appears almost linear. Although f is certainly not a linear function, we now investigate why the graph of f seems to be approaching a linear function. First, using long division of polynomials, we can write

$$f(x) = \frac{3x^2 + 4x}{x + 2} = 3x - 2 + \frac{4}{x + 2}.$$

Since $\frac{4}{(x+2)} \rightarrow 0$ as $x \rightarrow \pm\infty$, we conclude that

$$\lim_{x \rightarrow \pm\infty} (f(x) - (3x - 2)) = \lim_{x \rightarrow \pm\infty} \frac{4}{x + 2} = 0.$$

Therefore, the graph of f approaches the line $y = 3x - 2$ as $x \rightarrow \pm\infty$. This line is known as an **oblique asymptote** for f ([Figure 4.56](#)).

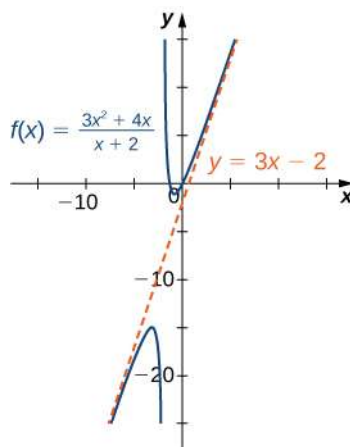


Figure 4.56 The graph of the rational function $f(x) = (3x^2 + 4x)/(x + 2)$ approaches the oblique asymptote $y = 3x - 2$ as $x \rightarrow \pm\infty$.

We can summarize the results of [Example 4.25](#) to make the following conclusion regarding end behavior for rational functions. Consider a rational function

$$f(x) = \frac{p(x)}{q(x)} = \frac{a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0}{b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0},$$

where $a_n \neq 0$ and $b_m \neq 0$.

1. If the degree of the numerator is the same as the degree of the denominator ($n = m$), then f has a horizontal asymptote of $y = a_n/b_m$ as $x \rightarrow \pm\infty$.
2. If the degree of the numerator is less than the degree of the denominator ($n < m$), then f has a horizontal asymptote of $y = 0$ as $x \rightarrow \pm\infty$.
3. If the degree of the numerator is greater than the degree of the denominator ($n > m$), then f does not have a horizontal asymptote. The limits at infinity are either positive or negative infinity, depending on the signs of the leading terms. In addition, using long division, the function can be rewritten as

$$f(x) = \frac{p(x)}{q(x)} = g(x) + \frac{r(x)}{q(x)},$$

where the degree of $r(x)$ is less than the degree of $q(x)$. As a result, $\lim_{x \rightarrow \pm\infty} r(x)/q(x) = 0$. Therefore, the values of

$[f(x) - g(x)]$ approach zero as $x \rightarrow \pm\infty$. If the degree of $p(x)$ is exactly one more than the degree of $q(x)$ ($n = m + 1$), the function $g(x)$ is a linear function. In this case, we call $g(x)$ an oblique asymptote. Now let's consider the end behavior for functions involving a radical.

EXAMPLE 4.26

Determining End Behavior for a Function Involving a Radical

Find the limits as $x \rightarrow \infty$ and $x \rightarrow -\infty$ for $f(x) = \frac{3x-2}{\sqrt{4x^2+5}}$ and describe the end behavior of f .

Solution

Let's use the same strategy as we did for rational functions: divide the numerator and denominator by a power of x . To determine the appropriate power of x , consider the expression $\sqrt{4x^2+5}$ in the denominator. Since

$$\sqrt{4x^2+5} \approx \sqrt{4x^2} = 2|x|$$

for large values of x in effect x appears just to the first power in the denominator. Therefore, we divide the numerator and denominator by $|x|$. Then, using the fact that $|x| = x$ for $x > 0$, $|x| = -x$ for $x < 0$, and $|x| = \sqrt{x^2}$ for all x , we calculate the limits as follows:

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \frac{3x-2}{\sqrt{4x^2+5}} &= \lim_{x \rightarrow \infty} \frac{(1/|x|)(3x-2)}{(1/|x|)\sqrt{4x^2+5}} \\
 &= \lim_{x \rightarrow \infty} \frac{(1/x)(3x-2)}{\sqrt{(1/x^2)(4x^2+5)}} \\
 &= \lim_{x \rightarrow \infty} \frac{3-2/x}{\sqrt{4+5/x^2}} = \frac{3}{\sqrt{4}} = \frac{3}{2} \\
 \lim_{x \rightarrow -\infty} \frac{3x-2}{\sqrt{4x^2+5}} &= \lim_{x \rightarrow -\infty} \frac{(1/|x|)(3x-2)}{(1/|x|)\sqrt{4x^2+5}} \\
 &= \lim_{x \rightarrow -\infty} \frac{(-1/x)(3x-2)}{\sqrt{(1/x^2)(4x^2+5)}} \\
 &= \lim_{x \rightarrow -\infty} \frac{-3+2/x}{\sqrt{4+5/x^2}} = \frac{-3}{\sqrt{4}} = -\frac{3}{2}.
 \end{aligned}$$

Therefore, $f(x)$ approaches the horizontal asymptote $y = \frac{3}{2}$ as $x \rightarrow \infty$ and the horizontal asymptote $y = -\frac{3}{2}$ as $x \rightarrow -\infty$ as shown in the following graph.

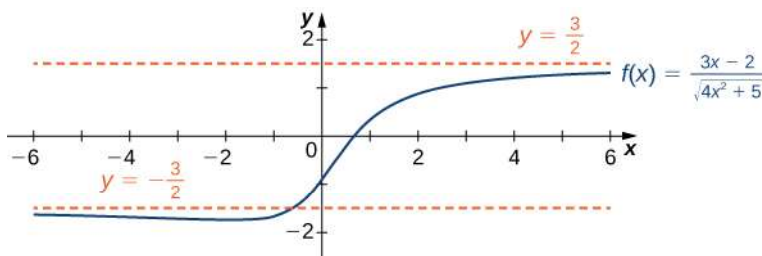


Figure 4.57 This function has two horizontal asymptotes and it crosses one of the asymptotes.

✓ 4.25 Evaluate $\lim_{x \rightarrow \infty} \frac{\sqrt{3x^2+4}}{x+6}$.

Determining End Behavior for Transcendental Functions

The six basic trigonometric functions are periodic and do not approach a finite limit as $x \rightarrow \pm\infty$. For example, $\sin x$ oscillates between 1 and -1 (Figure 4.58). The tangent function x has an infinite number of vertical asymptotes as $x \rightarrow \pm\infty$; therefore, it does not approach a finite limit nor does it approach $\pm\infty$ as $x \rightarrow \pm\infty$ as shown in Figure 4.59.

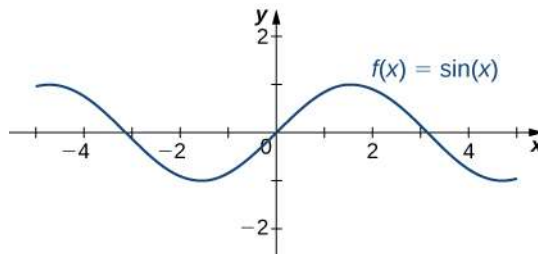


Figure 4.58 The function $f(x) = \sin x$ oscillates between 1 and -1 as $x \rightarrow \pm\infty$

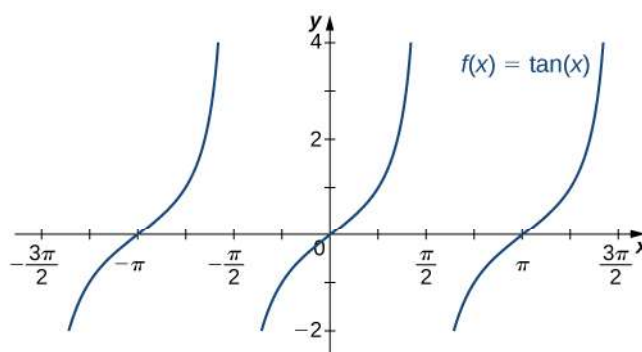


Figure 4.59 The function $f(x) = \tan x$ does not approach a limit and does not approach $\pm\infty$ as $x \rightarrow \pm\infty$

Recall that for any base $b > 0$, $b \neq 1$, the function $y = b^x$ is an exponential function with domain $(-\infty, \infty)$ and range $(0, \infty)$. If $b > 1$, $y = b^x$ is increasing over $(-\infty, \infty)$. If $0 < b < 1$, $y = b^x$ is decreasing over $(-\infty, \infty)$. For the natural exponential function $f(x) = e^x$, $e \approx 2.718 > 1$. Therefore, $f(x) = e^x$ is increasing on $(-\infty, \infty)$ and the range is $(0, \infty)$. The exponential function $f(x) = e^x$ approaches ∞ as $x \rightarrow \infty$ and approaches 0 as $x \rightarrow -\infty$ as shown in [Table 4.5](#) and [Figure 4.60](#).

x	-5	-2	0	2	5
e^x	0.00674	0.135	1	7.389	148.413

Table 4.5 End behavior of the natural exponential function

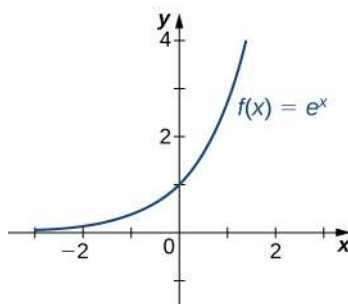


Figure 4.60 The exponential function approaches zero as $x \rightarrow -\infty$ and approaches ∞ as $x \rightarrow \infty$.

Recall that the natural logarithm function $f(x) = \ln(x)$ is the inverse of the natural exponential function $y = e^x$. Therefore, the domain of $f(x) = \ln(x)$ is $(0, \infty)$ and the range is $(-\infty, \infty)$. The graph of $f(x) = \ln(x)$ is the reflection of the graph of $y = e^x$ about the line $y = x$. Therefore, $\ln(x) \rightarrow -\infty$ as $x \rightarrow 0^+$ and $\ln(x) \rightarrow \infty$ as $x \rightarrow \infty$ as shown in [Figure 4.61](#) and [Table 4.6](#).

x	0.01	0.1	1	10	100
$\ln(x)$	-4.605	-2.303	0	2.303	4.605

Table 4.6 End behavior of the natural logarithm function

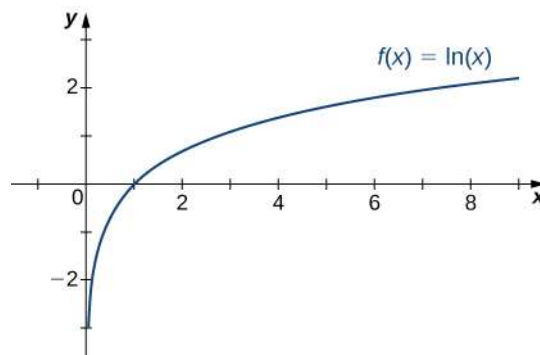


Figure 4.61 The natural logarithm function approaches ∞ as $x \rightarrow \infty$.

EXAMPLE 4.27

Determining End Behavior for a Transcendental Function

Find the limits as $x \rightarrow \infty$ and $x \rightarrow -\infty$ for $f(x) = \frac{(2+3e^x)}{(7-5e^x)}$ and describe the end behavior of f .

Solution

To find the limit as $x \rightarrow \infty$, divide the numerator and denominator by e^x :

$$\begin{aligned}\lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{2+3e^x}{7-5e^x} \\ &= \lim_{x \rightarrow \infty} \frac{(2/e^x)+3}{(7/e^x)-5}.\end{aligned}$$

As shown in [Figure 4.60](#), $e^x \rightarrow \infty$ as $x \rightarrow \infty$. Therefore,

$$\lim_{x \rightarrow \infty} \frac{2}{e^x} = 0 = \lim_{x \rightarrow \infty} \frac{7}{e^x}.$$

We conclude that $\lim_{x \rightarrow \infty} f(x) = -\frac{3}{5}$, and the graph of f approaches the horizontal asymptote $y = -\frac{3}{5}$ as $x \rightarrow \infty$. To find

the limit as $x \rightarrow -\infty$, use the fact that $e^x \rightarrow 0$ as $x \rightarrow -\infty$ to conclude that $\lim_{x \rightarrow -\infty} f(x) = \frac{2}{7}$, and therefore the graph of

approaches the horizontal asymptote $y = \frac{2}{7}$ as $x \rightarrow -\infty$.

 4.26 Find the limits as $x \rightarrow \infty$ and $x \rightarrow -\infty$ for $f(x) = \frac{(3e^x-4)}{(5e^x+2)}$.

Guidelines for Drawing the Graph of a Function

We now have enough analytical tools to draw graphs of a wide variety of algebraic and transcendental functions. Before showing how to graph specific functions, let's look at a general strategy to use when graphing any function.

Problem-Solving Strategy

Drawing the Graph of a Function

Given a function f , use the following steps to sketch a graph of f :

1. Determine the domain of the function.
2. Locate the x - and y -intercepts.

3. Evaluate $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$ to determine the end behavior. If either of these limits is a finite number L ,

then $y = L$ is a horizontal asymptote. If either of these limits is ∞ or $-\infty$, determine whether f has an oblique asymptote. If f is a rational function such that $f(x) = \frac{p(x)}{q(x)}$, where the degree of the numerator is greater than the degree of the denominator, then f can be written as

$$f(x) = \frac{p(x)}{q(x)} = g(x) + \frac{r(x)}{q(x)},$$

where the degree of $r(x)$ is less than the degree of $q(x)$. The values of $f(x)$ approach the values of $g(x)$ as $x \rightarrow \pm\infty$. If $g(x)$ is a linear function, it is known as an *oblique asymptote*.

4. Determine whether f has any vertical asymptotes.
5. Calculate f' . Find all critical points and determine the intervals where f is increasing and where f is decreasing. Determine whether f has any local extrema.
6. Calculate f'' . Determine the intervals where f is concave up and where f is concave down. Use this information to determine whether f has any inflection points. The second derivative can also be used as an alternate means to determine or verify that f has a local extremum at a critical point.

Now let's use this strategy to graph several different functions. We start by graphing a polynomial function.

EXAMPLE 4.28

Sketching a Graph of a Polynomial

Sketch a graph of $f(x) = (x - 1)^2(x + 2)$.

Solution

Step 1. Since f is a polynomial, the domain is the set of all real numbers.

Step 2. When $x = 0$, $f(x) = 2$. Therefore, the y -intercept is $(0, 2)$. To find the x -intercepts, we need to solve the equation $(x - 1)^2(x + 2) = 0$, gives us the x -intercepts $(1, 0)$ and $(-2, 0)$.

Step 3. We need to evaluate the end behavior of f . As $x \rightarrow \infty$, $(x - 1)^2 \rightarrow \infty$ and $(x + 2) \rightarrow \infty$. Therefore,

$\lim_{x \rightarrow \infty} f(x) = \infty$. As $x \rightarrow -\infty$, $(x - 1)^2 \rightarrow \infty$ and $(x + 2) \rightarrow -\infty$. Therefore, $\lim_{x \rightarrow -\infty} f(x) = -\infty$. To get even more

information about the end behavior of f , we can multiply the factors of f . When doing so, we see that

$$f(x) = (x - 1)^2(x + 2) = x^3 - 3x + 2.$$

Since the leading term of f is x^3 , we conclude that f behaves like $y = x^3$ as $x \rightarrow \pm\infty$.

Step 4. Since f is a polynomial function, it does not have any vertical asymptotes.

Step 5. The first derivative of f is

$$f'(x) = 3x^2 - 3.$$

Therefore, f has two critical points: $x = 1, -1$. Divide the interval $(-\infty, \infty)$ into the three smaller intervals: $(-\infty, -1)$, $(-1, 1)$, and $(1, \infty)$. Then, choose test points $x = -2$, $x = 0$, and $x = 2$ from these intervals and evaluate the sign of $f'(x)$ at each of these test points, as shown in the following table.

Interval	Test Point	Sign of Derivative $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$	Conclusion
$(-\infty, -1)$	$x = -2$	$(+)(-)(-) = +$	f is increasing.
$(-1, 1)$	$x = 0$	$(+)(-)(+) = -$	f is decreasing.
$(1, \infty)$	$x = 2$	$(+)(+)(+) = +$	f is increasing.

From the table, we see that f has a local maximum at $x = -1$ and a local minimum at $x = 1$. Evaluating $f(x)$ at those two points, we find that the local maximum value is $f(-1) = 4$ and the local minimum value is $f(1) = 0$.

Step 6. The second derivative of f is

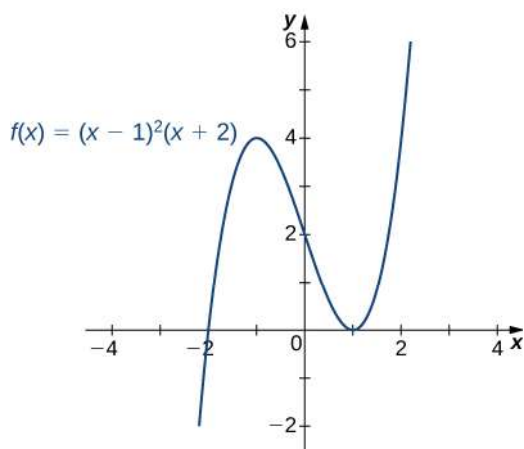
$$f''(x) = 6x.$$

The second derivative is zero at $x = 0$. Therefore, to determine the concavity of f , divide the interval $(-\infty, \infty)$ into the smaller intervals $(-\infty, 0)$ and $(0, \infty)$, and choose test points $x = -1$ and $x = 1$ to determine the concavity of f on each of these smaller intervals as shown in the following table.

Interval	Test Point	Sign of $f''(x) = 6x$	Conclusion
$(-\infty, 0)$	$x = -1$	$-$	f is concave down.
$(0, \infty)$	$x = 1$	$+$	f is concave up.

We note that the information in the preceding table confirms the fact, found in step 5, that f has a local maximum at $x = -1$ and a local minimum at $x = 1$. In addition, the information found in step 5—namely, f has a local maximum at $x = -1$ and a local minimum at $x = 1$, and $f'(x) = 0$ at those points—combined with the fact that f'' changes sign only at $x = 0$ confirms the results found in step 6 on the concavity of f .

Combining this information, we arrive at the graph of $f(x) = (x-1)^2(x+2)$ shown in the following graph.



☒ 4.27 Sketch a graph of $f(x) = (x-1)^3(x+2)$.

EXAMPLE 4.29**Sketching a Rational Function**

Sketch the graph of $f(x) = \frac{x^2}{(1-x^2)}$.

✓ Solution

Step 1. The function f is defined as long as the denominator is not zero. Therefore, the domain is the set of all real numbers x except $x = \pm 1$.

Step 2. Find the intercepts. If $x = 0$, then $f(x) = 0$, so 0 is an intercept. If $y = 0$, then $\frac{x^2}{(1-x^2)} = 0$, which implies $x = 0$.

Therefore, $(0, 0)$ is the only intercept.

Step 3. Evaluate the limits at infinity. Since f is a rational function, divide the numerator and denominator by the highest power in the denominator: x^2 . We obtain

$$\lim_{x \rightarrow \pm\infty} \frac{x^2}{1-x^2} = \lim_{x \rightarrow \pm\infty} \frac{1}{\frac{1}{x^2} - 1} = -1.$$

Therefore, f has a horizontal asymptote of $y = -1$ as $x \rightarrow \infty$ and $x \rightarrow -\infty$.

Step 4. To determine whether f has any vertical asymptotes, first check to see whether the denominator has any zeroes. We find the denominator is zero when $x = \pm 1$. To determine whether the lines $x = 1$ or $x = -1$ are vertical asymptotes of f , evaluate $\lim_{x \rightarrow 1} f(x)$ and $\lim_{x \rightarrow -1} f(x)$. By looking at each one-sided limit as $x \rightarrow 1$, we see that

$$\lim_{x \rightarrow 1^+} \frac{x^2}{1-x^2} = -\infty \text{ and } \lim_{x \rightarrow 1^-} \frac{x^2}{1-x^2} = \infty.$$

In addition, by looking at each one-sided limit as $x \rightarrow -1$, we find that

$$\lim_{x \rightarrow -1^+} \frac{x^2}{1-x^2} = \infty \text{ and } \lim_{x \rightarrow -1^-} \frac{x^2}{1-x^2} = -\infty.$$

Step 5. Calculate the first derivative:

$$f'(x) = \frac{(1-x^2)(2x) - x^2(-2x)}{(1-x^2)^2} = \frac{2x}{(1-x^2)^2}.$$

Critical points occur at points x where $f'(x) = 0$ or $f'(x)$ is undefined. We see that $f'(x) = 0$ when $x = 0$. The derivative f' is not undefined at any point in the domain of f . However, $x = \pm 1$ are not in the domain of f . Therefore, to determine where f is increasing and where f is decreasing, divide the interval $(-\infty, \infty)$ into four smaller intervals: $(-\infty, -1)$, $(-1, 0)$, $(0, 1)$, and $(1, \infty)$, and choose a test point in each interval to determine the sign of $f'(x)$ in each of these intervals. The values $x = -2$, $x = -\frac{1}{2}$, $x = \frac{1}{2}$, and $x = 2$ are good choices for test points as shown in the following table.

Interval	Test Point	Sign of $f'(x) = \frac{2x}{(1-x^2)^2}$	Conclusion
$(-\infty, -1)$	$x = -2$	$-/+ = -$	f is decreasing.
$(-1, 0)$	$x = -1/2$	$-/+ = -$	f is decreasing.

Interval	Test Point	Sign of $f'(x) = \frac{2x}{(1-x^2)^2}$	Conclusion
$(0, 1)$	$x = 1/2$	$+/+ = +$	f is increasing.
$(1, \infty)$	$x = 2$	$+/+ = +$	f is increasing.

From this analysis, we conclude that f has a local minimum at $x = 0$ but no local maximum.

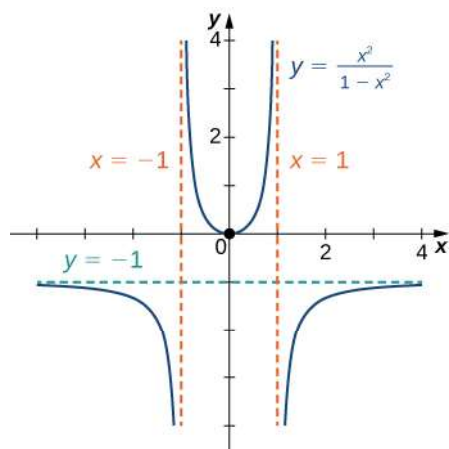
Step 6. Calculate the second derivative:

$$\begin{aligned}
 f''(x) &= \frac{(1-x^2)^2(2) - 2x(2(1-x^2)(-2x))}{(1-x^2)^4} \\
 &= \frac{(1-x^2)[2(1-x^2) + 8x^2]}{(1-x^2)^4} \\
 &= \frac{2(1-x^2) + 8x^2}{(1-x^2)^3} \\
 &= \frac{6x^2 + 2}{(1-x^2)^3}.
 \end{aligned}$$

To determine the intervals where f is concave up and where f is concave down, we first need to find all points x where $f''(x) = 0$ or $f''(x)$ is undefined. Since the numerator $6x^2 + 2 \neq 0$ for any x , $f''(x)$ is never zero. Furthermore, f'' is not undefined for any x in the domain of f . However, as discussed earlier, $x = \pm 1$ are not in the domain of f . Therefore, to determine the concavity of f , we divide the interval $(-\infty, \infty)$ into the three smaller intervals $(-\infty, -1)$, $(-1, 1)$, and $(1, \infty)$, and choose a test point in each of these intervals to evaluate the sign of $f''(x)$ in each of these intervals. The values $x = -2$, $x = 0$, and $x = 2$ are possible test points as shown in the following table.

Interval	Test Point	Sign of $f''(x) = \frac{6x^2+2}{(1-x^2)^3}$	Conclusion
$(-\infty, -1)$	$x = -2$	$+/- = -$	f is concave down.
$(-1, 1)$	$x = 0$	$+/+ = +$	f is concave up.
$(1, \infty)$	$x = 2$	$+/- = -$	f is concave down.

Combining all this information, we arrive at the graph of f shown below. Note that, although f changes concavity at $x = -1$ and $x = 1$, there are no inflection points at either of these places because f is not continuous at $x = -1$ or $x = 1$.



- ✓ 4.28 Sketch a graph of $f(x) = \frac{(3x+5)}{(8+4x)}$.

EXAMPLE 4.30

Sketching a Rational Function with an Oblique Asymptote

Sketch the graph of $f(x) = \frac{x^2}{(x-1)}$

✓ Solution

Step 1. The domain of f is the set of all real numbers x except $x = 1$.

Step 2. Find the intercepts. We can see that when $x = 0$, $f(x) = 0$, so $(0, 0)$ is the only intercept.

Step 3. Evaluate the limits at infinity. Since the degree of the numerator is one more than the degree of the denominator, f must have an oblique asymptote. To find the oblique asymptote, use long division of polynomials to write

$$f(x) = \frac{x^2}{x-1} = x + 1 + \frac{1}{x-1}.$$

Since $1/(x-1) \rightarrow 0$ as $x \rightarrow \pm\infty$, $f(x)$ approaches the line $y = x + 1$ as $x \rightarrow \pm\infty$. The line $y = x + 1$ is an oblique asymptote for f .

Step 4. To check for vertical asymptotes, look at where the denominator is zero. Here the denominator is zero at $x = 1$. Looking at both one-sided limits as $x \rightarrow 1$, we find

$$\lim_{x \rightarrow 1^+} \frac{x^2}{x-1} = \infty \text{ and } \lim_{x \rightarrow 1^-} \frac{x^2}{x-1} = -\infty.$$

Therefore, $x = 1$ is a vertical asymptote, and we have determined the behavior of f as x approaches 1 from the right and the left.

Step 5. Calculate the first derivative:

$$f'(x) = \frac{(x-1)(2x) - x^2(1)}{(x-1)^2} = \frac{x^2 - 2x}{(x-1)^2}.$$

We have $f'(x) = 0$ when $x^2 - 2x = x(x-2) = 0$. Therefore, $x = 0$ and $x = 2$ are critical points. Since f is undefined at $x = 1$, we need to divide the interval $(-\infty, \infty)$ into the smaller intervals $(-\infty, 0)$, $(0, 1)$, $(1, 2)$, and $(2, \infty)$, and choose a test point from each interval to evaluate the sign of $f'(x)$ in each of these smaller intervals. For example, let $x = -1$, $x = \frac{1}{2}$, $x = \frac{3}{2}$, and $x = 3$ be the test points as shown in the following table.

Interval	Test Point	Sign of $f'(x) = \frac{x^2-2x}{(x-1)^2} = \frac{x(x-2)}{(x-1)^2}$	Conclusion
$(-\infty, 0)$	$x = -1$	$(-)(-)/+ = +$	f is increasing.
$(0, 1)$	$x = 1/2$	$(+)(-)/+ = -$	f is decreasing.
$(1, 2)$	$x = 3/2$	$(+)(-)/+ = -$	f is decreasing.
$(2, \infty)$	$x = 3$	$(+)(+)/+ = +$	f is increasing.

From this table, we see that f has a local maximum at $x = 0$ and a local minimum at $x = 2$. The value of f at the local maximum is $f(0) = 0$ and the value of f at the local minimum is $f(2) = 4$. Therefore, $(0, 0)$ and $(2, 4)$ are important points on the graph.

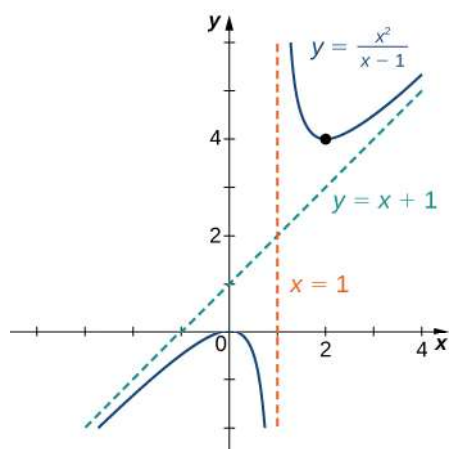
Step 6. Calculate the second derivative:

$$\begin{aligned}
 f''(x) &= \frac{(x-1)^2(2x-2) - (x^2-2x)(2(x-1))}{(x-1)^4} \\
 &= \frac{(x-1)[(x-1)(2x-2) - 2(x^2-2x)]}{(x-1)^4} \\
 &= \frac{(x-1)(2x-2) - 2(x^2-2x)}{(x-1)^3} \\
 &= \frac{2x^2 - 4x + 2 - (2x^2 - 4x)}{(x-1)^3} \\
 &= \frac{2}{(x-1)^3}.
 \end{aligned}$$

We see that $f''(x)$ is never zero or undefined for x in the domain of f . Since f is undefined at $x = 1$, to check concavity we just divide the interval $(-\infty, \infty)$ into the two smaller intervals $(-\infty, 1)$ and $(1, \infty)$, and choose a test point from each interval to evaluate the sign of $f''(x)$ in each of these intervals. The values $x = 0$ and $x = 2$ are possible test points as shown in the following table.

Interval	Test Point	Sign of $f''(x) = \frac{2}{(x-1)^3}$	Conclusion
$(-\infty, 1)$	$x = 0$	$+/- = -$	f is concave down.
$(1, \infty)$	$x = 2$	$+/+ = +$	f is concave up.

From the information gathered, we arrive at the following graph for f .



- ✓ 4.29 Find the oblique asymptote for $f(x) = \frac{(3x^3 - 2x + 1)}{(2x^2 - 4)}$.

EXAMPLE 4.31

Sketching the Graph of a Function with a Cusp

Sketch a graph of $f(x) = (x - 1)^{2/3}$.

✓ Solution

Step 1: Since the cube-root function is defined for all real numbers x and $(x - 1)^{2/3} = \left(\sqrt[3]{x - 1}\right)^2$, the domain of f is all real numbers.

Step 2: To find the y -intercept, evaluate $f(0)$. Since $f(0) = 1$, the y -intercept is $(0, 1)$. To find the x -intercept, solve $(x - 1)^{2/3} = 0$. The solution of this equation is $x = 1$, so the x -intercept is $(1, 0)$.

Step 3: Since $\lim_{x \rightarrow \pm\infty} (x - 1)^{2/3} = \infty$, the function continues to grow without bound as $x \rightarrow \infty$ and $x \rightarrow -\infty$.

Step 4: The function has no vertical asymptotes.

Step 5: To determine where f is increasing or decreasing, calculate f' . We find

$$f'(x) = \frac{2}{3}(x - 1)^{-1/3} = \frac{2}{3(x - 1)^{1/3}}.$$

This function is not zero anywhere, but it is undefined when $x = 1$. Therefore, the only critical point is $x = 1$. Divide the interval $(-\infty, \infty)$ into the smaller intervals $(-\infty, 1)$ and $(1, \infty)$, and choose test points in each of these intervals to determine the sign of $f'(x)$ in each of these smaller intervals. Let $x = 0$ and $x = 2$ be the test points as shown in the following table.

Interval	Test Point	Sign of $f'(x) = \frac{2}{3(x-1)^{1/3}}$	Conclusion
$(-\infty, 1)$	$x = 0$	$+/- = -$	f is decreasing.
$(1, \infty)$	$x = 2$	$+/+ = +$	f is increasing.

We conclude that f has a local minimum at $x = 1$. Evaluating f at $x = 1$, we find that the value of f at the local

minimum is zero. Note that $f'(1)$ is undefined, so to determine the behavior of the function at this critical point, we need to examine $\lim_{x \rightarrow 1} f'(x)$. Looking at the one-sided limits, we have

$$\lim_{x \rightarrow 1^+} \frac{2}{3(x-1)^{1/3}} = \infty \text{ and } \lim_{x \rightarrow 1^-} \frac{2}{3(x-1)^{1/3}} = -\infty.$$

Therefore, f has a cusp at $x = 1$.

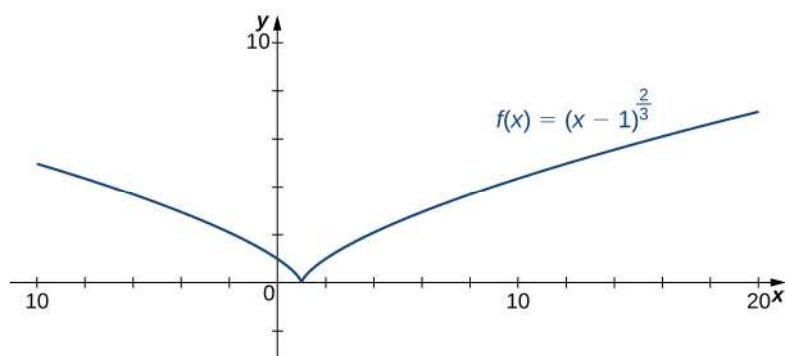
Step 6: To determine concavity, we calculate the second derivative of f :

$$f''(x) = -\frac{2}{9}(x-1)^{-4/3} = \frac{-2}{9(x-1)^{4/3}}.$$

We find that $f''(x)$ is defined for all x , but is undefined when $x = 1$. Therefore, divide the interval $(-\infty, \infty)$ into the smaller intervals $(-\infty, 1)$ and $(1, \infty)$, and choose test points to evaluate the sign of $f''(x)$ in each of these intervals. As we did earlier, let $x = 0$ and $x = 2$ be test points as shown in the following table.

Interval	Test Point	Sign of $f''(x) = \frac{-2}{9(x-1)^{4/3}}$	Conclusion
$(-\infty, 1)$	$x = 0$	$-/+ = -$	f is concave down.
$(1, \infty)$	$x = 2$	$-/+ = -$	f is concave down.

From this table, we conclude that f is concave down everywhere. Combining all of this information, we arrive at the following graph for f .



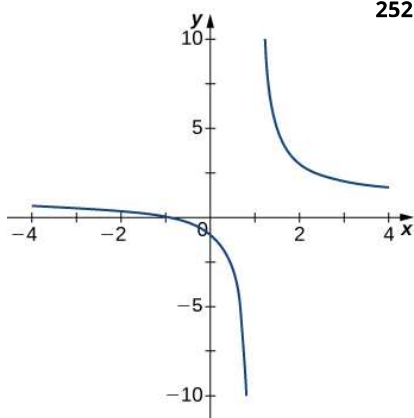
- ✓ 4.30 Consider the function $f(x) = 5 - x^{2/3}$. Determine the point on the graph where a cusp is located. Determine the end behavior of f .



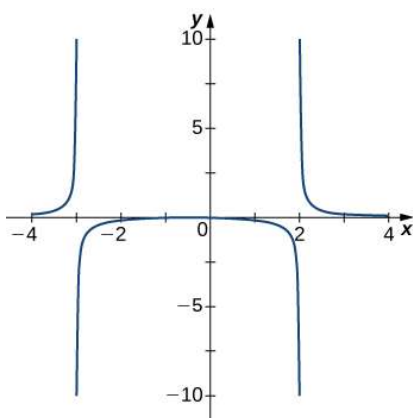
SECTION 4.6 EXERCISES

For the following exercises, examine the graphs. Identify where the vertical asymptotes are located.

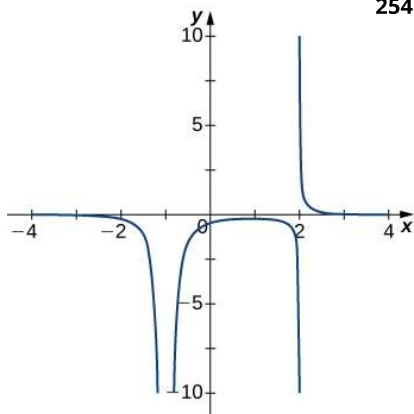
251.



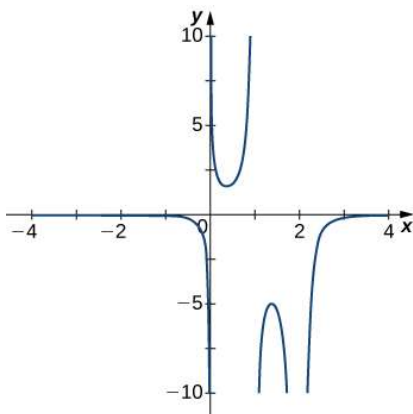
252.



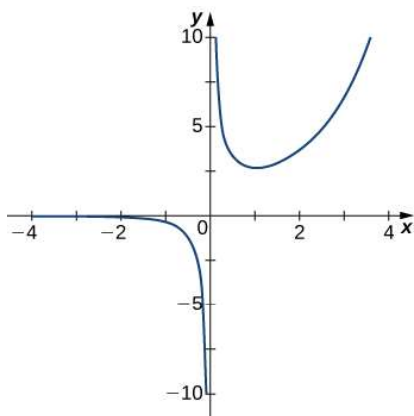
253.



254.



255.



For the following functions $f(x)$, determine whether there is an asymptote at $x = a$. Justify your answer without graphing on a calculator.

256. $f(x) = \frac{x+1}{x^2+5x+4}, a = -1$

257. $f(x) = \frac{x}{x-2}, a = 2$

258. $f(x) = (x+2)^{3/2}, a = -2$

259. $f(x) = (x-1)^{-1/3}, a = 1$

260. $f(x) = 1 + x^{-2/5}, a = 1$

For the following exercises, evaluate the limit.

$$261. \lim_{x \rightarrow \infty} \frac{1}{3x+6}$$

$$262. \lim_{x \rightarrow \infty} \frac{2x-5}{4x}$$

$$263. \lim_{x \rightarrow \infty} \frac{x^2-2x+5}{x+2}$$

$$264. \lim_{x \rightarrow -\infty} \frac{3x^3-2x}{x^2+2x+8}$$

$$265. \lim_{x \rightarrow -\infty} \frac{x^4-4x^3+1}{2-2x^2-7x^4}$$

$$266. \lim_{x \rightarrow \infty} \frac{3x}{\sqrt{x^2+1}}$$

$$267. \lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2-1}}{x+2}$$

$$268. \lim_{x \rightarrow \infty} \frac{4x}{\sqrt{x^2-1}}$$

$$269. \lim_{x \rightarrow -\infty} \frac{4x}{\sqrt{x^2-1}}$$

$$270. \lim_{x \rightarrow \infty} \frac{2\sqrt{x}}{x-\sqrt{x+1}}$$

For the following exercises, find the horizontal and vertical asymptotes.

$$271. f(x) = x - \frac{9}{x}$$

$$272. f(x) = \frac{1}{1-x^2}$$

$$273. f(x) = \frac{x^3}{4-x^2}$$

$$274. f(x) = \frac{x^2+3}{x^2+1}$$

$$275. f(x) = \sin(x) \sin(2x)$$

$$276. f(x) = \cos x + \cos(3x) + \cos(5x)$$

$$277. f(x) = \frac{x \sin(x)}{x^2-1}$$

$$278. f(x) = \frac{x}{\sin(x)}$$

$$279. f(x) = \frac{1}{x^3+x^2}$$

$$280. f(x) = \frac{1}{x-1} - 2x$$

$$281. f(x) = \frac{x^3+1}{x^3-1}$$

$$282. f(x) = \frac{\sin x + \cos x}{\sin x - \cos x}$$

$$283. f(x) = x - \sin x$$

$$284. f(x) = \frac{1}{x} - \sqrt{x}$$

For the following exercises, construct a function $f(x)$ that has the given asymptotes.

$$285. x = 1 \text{ and } y = 2$$

$$286. x = 1 \text{ and } y = 0$$

$$287. y = 4, x = -1$$

$$288. x = 0$$

For the following exercises, graph the function on a graphing calculator on the window $x = [-5, 5]$ and estimate the horizontal asymptote or limit. Then, calculate the actual horizontal asymptote or limit.

$$289. [T] f(x) = \frac{1}{x+10}$$

$$290. [T] f(x) = \frac{x+1}{x^2+7x+6}$$

$$291. [T] \lim_{x \rightarrow -\infty} x^2 + 10x + 25$$

$$292. [T] \lim_{x \rightarrow -\infty} \frac{x+2}{x^2+7x+6}$$

$$293. [T] \lim_{x \rightarrow \infty} \frac{3x+2}{x+5}$$

For the following exercises, draw a graph of the functions without using a calculator. Be sure to notice all important features of the graph: local maxima and minima, inflection points, and asymptotic behavior.

$$294. y = 3x^2 + 2x + 4$$

$$295. y = x^3 - 3x^2 + 4$$

$$296. y = \frac{2x+1}{x^2+6x+5}$$

$$297. y = \frac{x^3+4x^2+3x}{3x+9}$$

$$298. y = \frac{x^2+x-2}{x^2-3x-4}$$

$$299. y = \sqrt{x^2 - 5x + 4}$$

$$300. y = 2x\sqrt{16-x^2}$$

$$301. y = \frac{\cos x}{x}, \text{ on } x = [-2\pi, 2\pi]$$

$$302. y = \frac{\sqrt{x^2+2}}{x+1}$$

$$303. y = x \tan x, x = [-\pi, \pi]$$

$$304. y = x \ln(x), x > 0$$

$$305. y = x^2 \sin(x), x = [-2\pi, 2\pi]$$

$$306. \text{ For } f(x) = \frac{P(x)}{Q(x)} \text{ to have an asymptote at } y = 2 \text{ then the polynomials } P(x) \text{ and } Q(x) \text{ must have what relation?}$$

$$307. \text{ For } f(x) = \frac{P(x)}{Q(x)} \text{ to have an asymptote at } x = 0, \text{ then the polynomials } P(x) \text{ and } Q(x) \text{ must have what relation?}$$

$$308. \text{ If } f'(x) \text{ has asymptotes at } y = 3 \text{ and } x = 1, \text{ then } f(x) \text{ has what asymptotes?}$$

309. Both $f(x) = \frac{1}{(x-1)}$ and $g(x) = \frac{1}{(x-1)^2}$ have asymptotes at $x = 1$ and $y = 0$. What is the most obvious difference between these two functions?
310. True or false: Every ratio of polynomials has vertical asymptotes.

4.7 Applied Optimization Problems

Learning Objectives

4.7.1 Set up and solve optimization problems in several applied fields.

One common application of calculus is calculating the minimum or maximum value of a function. For example, companies often want to minimize production costs or maximize revenue. In manufacturing, it is often desirable to minimize the amount of material used to package a product with a certain volume. In this section, we show how to set up these types of minimization and maximization problems and solve them by using the tools developed in this chapter.

Solving Optimization Problems over a Closed, Bounded Interval

The basic idea of the **optimization problems** that follow is the same. We have a particular quantity that we are interested in maximizing or minimizing. However, we also have some auxiliary condition that needs to be satisfied. For example, in [Example 4.32](#), we are interested in maximizing the area of a rectangular garden. Certainly, if we keep making the side lengths of the garden larger, the area will continue to become larger. However, what if we have some restriction on how much fencing we can use for the perimeter? In this case, we cannot make the garden as large as we like. Let's look at how we can maximize the area of a rectangle subject to some constraint on the perimeter.

EXAMPLE 4.32

Maximizing the Area of a Garden

A rectangular garden is to be constructed using a rock wall as one side of the garden and wire fencing for the other three sides ([Figure 4.62](#)). Given 100 ft of wire fencing, determine the dimensions that would create a garden of maximum area. What is the maximum area?

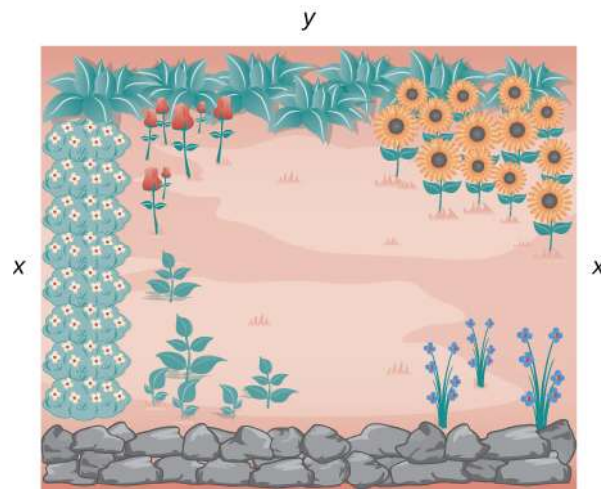


Figure 4.62 We want to determine the measurements x and y that will create a garden with a maximum area using 100 ft of fencing.

✓ Solution

Let x denote the length of the side of the garden perpendicular to the rock wall and y denote the length of the side parallel to the rock wall. Then the area of the garden is

$$A = x \cdot y.$$

We want to find the maximum possible area subject to the constraint that the total fencing is 100 ft. From [Figure 4.62](#), the total amount of fencing used will be $2x + y$. Therefore, the constraint equation is

$$2x + y = 100.$$

Solving this equation for y , we have $y = 100 - 2x$. Thus, we can write the area as

$$A(x) = x \cdot (100 - 2x) = 100x - 2x^2.$$

Before trying to maximize the area function $A(x) = 100x - 2x^2$, we need to determine the domain under consideration. To construct a rectangular garden, we certainly need the lengths of both sides to be positive. Therefore, we need $x > 0$ and $y > 0$. Since $y = 100 - 2x$, if $y > 0$, then $x < 50$. Therefore, we are trying to determine the maximum value of $A(x)$ for x over the open interval $(0, 50)$. We do not know that a function necessarily has a maximum value over an open interval. However, we do know that a continuous function has an absolute maximum (and absolute minimum) over a closed interval. Therefore, let's consider the function $A(x) = 100x - 2x^2$ over the closed interval $[0, 50]$. If the maximum value occurs at an interior point, then we have found the value x in the open interval $(0, 50)$ that maximizes the area of the garden. Therefore, we consider the following problem:

Maximize $A(x) = 100x - 2x^2$ over the interval $[0, 50]$.

As mentioned earlier, since A is a continuous function on a closed, bounded interval, by the extreme value theorem, it has a maximum and a minimum. These extreme values occur either at endpoints or critical points. At the endpoints, $A(x) = 0$. Since the area is positive for all x in the open interval $(0, 50)$, the maximum must occur at a critical point. Differentiating the function $A(x)$, we obtain

$$A'(x) = 100 - 4x.$$

Therefore, the only critical point is $x = 25$ (Figure 4.63). We conclude that the maximum area must occur when $x = 25$. Then we have $y = 100 - 2x = 100 - 2(25) = 50$. To maximize the area of the garden, let $x = 25$ ft and $y = 50$ ft. The area of this garden is 1250 ft^2 .

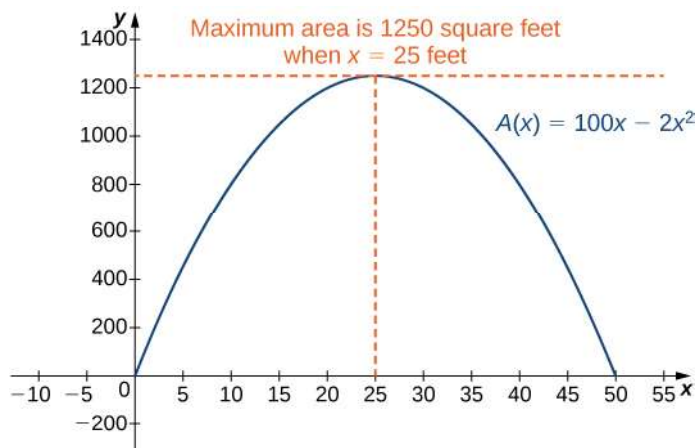


Figure 4.63 To maximize the area of the garden, we need to find the maximum value of the function $A(x) = 100x - 2x^2$.

- ✓ 4.31 Determine the maximum area if we want to make the same rectangular garden as in Figure 4.63, but we have 200 ft of fencing.

Now let's look at a general strategy for solving optimization problems similar to Example 4.32.

Problem-Solving Strategy

Solving Optimization Problems

1. Introduce all variables. If applicable, draw a figure and label all variables.
2. Determine which quantity is to be maximized or minimized, and for what range of values of the other variables (if this can be determined at this time).
3. Write a formula for the quantity to be maximized or minimized in terms of the variables. This formula may involve more than one variable.

4. Write any equations relating the independent variables in the formula from step 3. Use these equations to write the quantity to be maximized or minimized as a function of one variable.
5. Identify the domain of consideration for the function in step 4 based on the physical problem to be solved.
6. Locate the maximum or minimum value of the function from step 4. This step typically involves looking for critical points and evaluating a function at endpoints.

Now let's apply this strategy to maximize the volume of an open-top box given a constraint on the amount of material to be used.

EXAMPLE 4.33

Maximizing the Volume of a Box

An open-top box is to be made from a 24 in. by 36 in. piece of cardboard by removing a square from each corner of the box and folding up the flaps on each side. What size square should be cut out of each corner to get a box with the maximum volume?

✓ Solution

Step 1: Let x be the side length of the square to be removed from each corner (Figure 4.64). Then, the remaining four flaps can be folded up to form an open-top box. Let V be the volume of the resulting box.

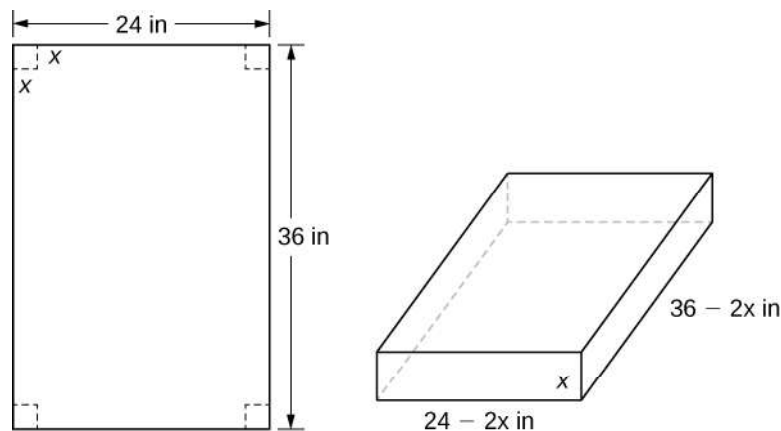


Figure 4.64 A square with side length x inches is removed from each corner of the piece of cardboard. The remaining flaps are folded to form an open-top box.

Step 2: We are trying to maximize the volume of a box. Therefore, the problem is to maximize V .

Step 3: As mentioned in step 2, we are trying to maximize the volume of a box. The volume of a box is $V = L \cdot W \cdot H$, where L , W , and H are the length, width, and height, respectively.

Step 4: From Figure 4.64, we see that the height of the box is x inches, the length is $36 - 2x$ inches, and the width is $24 - 2x$ inches. Therefore, the volume of the box is

$$V(x) = (36 - 2x)(24 - 2x)x = 4x^3 - 120x^2 + 864x.$$

Step 5: To determine the domain of consideration, let's examine Figure 4.64. Certainly, we need $x > 0$. Furthermore, the side length of the square cannot be greater than or equal to half the length of the shorter side, 24 in.; otherwise, one of the flaps would be completely cut off. Therefore, we are trying to determine whether there is a maximum volume of the box for x over the open interval $(0, 12)$. Since V is a continuous function over the closed interval $[0, 12]$, we know V will have an absolute maximum over the closed interval. Therefore, we consider V over the closed interval $[0, 12]$ and check whether the absolute maximum occurs at an interior point.

Step 6: Since $V(x)$ is a continuous function over the closed, bounded interval $[0, 12]$, V must have an absolute maximum (and an absolute minimum). Since $V(x) = 0$ at the endpoints and $V(x) > 0$ for $0 < x < 12$, the maximum must occur at a critical point. The derivative is

$$V'(x) = 12x^2 - 240x + 864.$$

To find the critical points, we need to solve the equation

$$12x^2 - 240x + 864 = 0.$$

Dividing both sides of this equation by 12, the problem simplifies to solving the equation

$$x^2 - 20x + 72 = 0.$$

Using the quadratic formula, we find that the critical points are

$$x = \frac{20 \pm \sqrt{(-20)^2 - 4(1)(72)}}{2} = \frac{20 \pm \sqrt{112}}{2} = \frac{20 \pm 4\sqrt{7}}{2} = 10 \pm 2\sqrt{7}.$$

Since $10 + 2\sqrt{7}$ is not in the domain of consideration, the only critical point we need to consider is $10 - 2\sqrt{7}$. Therefore, the volume is maximized if we let $x = 10 - 2\sqrt{7}$ in. The maximum volume is

$V(10 - 2\sqrt{7}) = 640 + 448\sqrt{7} \approx 1825 \text{ in.}^3$ as shown in the following graph.

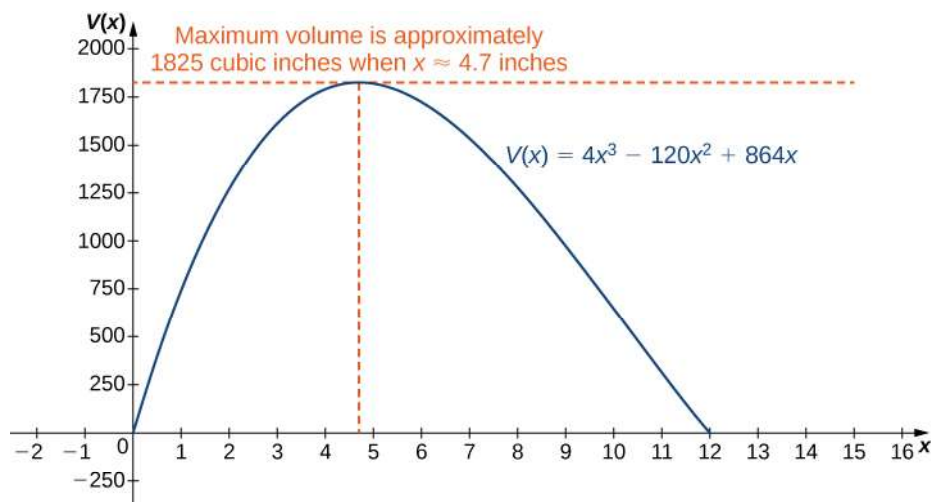


Figure 4.65 Maximizing the volume of the box leads to finding the maximum value of a cubic polynomial.

MEDIA

Watch a [video \(http://www.openstax.org/l/20_boxvolume\)](http://www.openstax.org/l/20_boxvolume) about optimizing the volume of a box.

- ☒ 4.32 Suppose the dimensions of the cardboard in [Example 4.33](#) are 20 in. by 30 in. Let x be the side length of each square and write the volume of the open-top box as a function of x . Determine the domain of consideration for x .

EXAMPLE 4.34

Minimizing Travel Time

An island is 2 mi due north of its closest point along a straight shoreline. A visitor is staying at a cabin on the shore that is 6 mi west of that point. The visitor is planning to go from the cabin to the island. Suppose the visitor runs at a rate of 8 mph and swims at a rate of 3 mph. How far should the visitor run before swimming to minimize the time it takes to reach the island?

Solution

Step 1: Let x be the distance running and let y be the distance swimming ([Figure 4.66](#)). Let T be the time it takes to get from the cabin to the island.

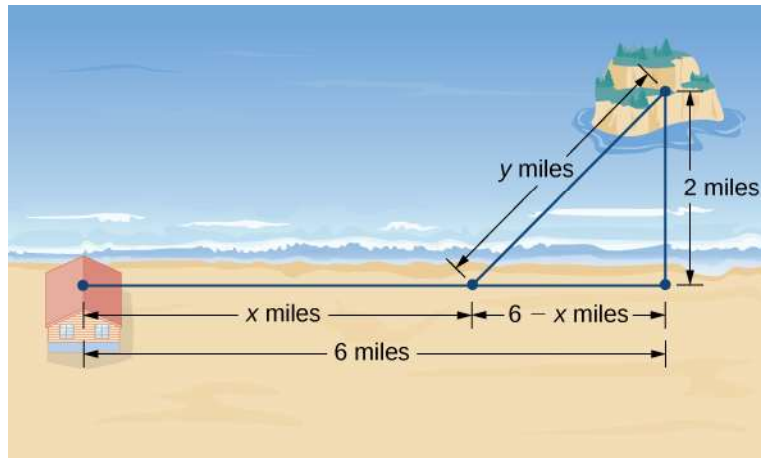


Figure 4.66 How can we choose x and y to minimize the travel time from the cabin to the island?

Step 2: The problem is to minimize T .

Step 3: To find the time spent traveling from the cabin to the island, add the time spent running and the time spent swimming. Since Distance = Rate $\times$ Time ($D = R \times T$), the time spent running is

$$T_{\text{running}} = \frac{D_{\text{running}}}{R_{\text{running}}} = \frac{x}{8},$$

and the time spent swimming is

$$T_{\text{swimming}} = \frac{D_{\text{swimming}}}{R_{\text{swimming}}} = \frac{y}{3}.$$

Therefore, the total time spent traveling is

$$T = \frac{x}{8} + \frac{y}{3}.$$

Step 4: From [Figure 4.66](#), the line segment of y miles forms the hypotenuse of a right triangle with legs of length 2 mi and $6 - x$ mi. Therefore, by the Pythagorean theorem, $2^2 + (6 - x)^2 = y^2$, and we obtain $y = \sqrt{(6 - x)^2 + 4}$. Thus, the total time spent traveling is given by the function

$$T(x) = \frac{x}{8} + \frac{\sqrt{(6 - x)^2 + 4}}{3}.$$

Step 5: From [Figure 4.66](#), we see that $0 \leq x \leq 6$. Therefore, $[0, 6]$ is the domain of consideration.

Step 6: Since $T(x)$ is a continuous function over a closed, bounded interval, it has a maximum and a minimum. Let's begin by looking for any critical points of T over the interval $[0, 6]$. The derivative is

$$T'(x) = \frac{1}{8} - \frac{1}{2} \frac{[(6 - x)^2 + 4]^{-1/2}}{3} \cdot 2(6 - x) = \frac{1}{8} - \frac{(6 - x)}{3\sqrt{(6 - x)^2 + 4}}.$$

If $T'(x) = 0$, then

$$\frac{1}{8} = \frac{6 - x}{3\sqrt{(6 - x)^2 + 4}}.$$

Therefore,

$$3\sqrt{(6 - x)^2 + 4} = 8(6 - x). \quad (4.6)$$

Squaring both sides of this equation, we see that if x satisfies this equation, then x must satisfy

$$9[(6 - x)^2 + 4] = 64(6 - x)^2,$$

which implies

$$55(6 - x)^2 = 36.$$

We conclude that if x is a critical point, then x satisfies

$$(x - 6)^2 = \frac{36}{55}.$$

Therefore, the possibilities for critical points are

$$x = 6 \pm \frac{6}{\sqrt{55}}.$$

Since $x = 6 + 6/\sqrt{55}$ is not in the domain, it is not a possibility for a critical point. On the other hand, $x = 6 - 6/\sqrt{55}$ is in the domain. Since we squared both sides of Equation 4.6 to arrive at the possible critical points, it remains to verify that $x = 6 - 6/\sqrt{55}$ satisfies Equation 4.6. Since $x = 6 - 6/\sqrt{55}$ does satisfy that equation, we conclude that $x = 6 - 6/\sqrt{55}$ is a critical point, and it is the only one. To justify that the time is minimized for this value of x , we just need to check the values of $T(x)$ at the endpoints $x = 0$ and $x = 6$, and compare them with the value of $T(x)$ at the critical point $x = 6 - 6/\sqrt{55}$. We find that $T(0) \approx 2.108$ h and $T(6) \approx 1.417$ h, whereas $T(6 - 6/\sqrt{55}) \approx 1.368$ h. Therefore, we conclude that T has a local minimum at $x \approx 5.19$ mi.

- ✓ 4.33 Suppose the island is 1 mi from shore, and the distance from the cabin to the point on the shore closest to the island is 15 mi. Suppose a visitor swims at the rate of 2.5 mph and runs at a rate of 6 mph. Let x denote the distance the visitor will run before swimming, and find a function for the time it takes the visitor to get from the cabin to the island.

In business, companies are interested in maximizing revenue. In the following example, we consider a scenario in which a company has collected data on how many cars it is able to lease, depending on the price it charges its customers to rent a car. Let's use these data to determine the price the company should charge to maximize the amount of money it brings in.

EXAMPLE 4.35

Maximizing Revenue

Owners of a car rental company have determined that if they charge customers p dollars per day to rent a car, where $50 \leq p \leq 200$, the number of cars n they rent per day can be modeled by the linear function $n(p) = 1000 - 5p$. If they charge \$50 per day or less, they will rent all their cars. If they charge \$200 per day or more, they will not rent any cars. Assuming the owners plan to charge customers between \$50 per day and \$200 per day to rent a car, how much should they charge to maximize their revenue?

Solution

Step 1: Let p be the price charged per car per day and let n be the number of cars rented per day. Let R be the revenue per day.

Step 2: The problem is to maximize R .

Step 3: The revenue (per day) is equal to the number of cars rented per day times the price charged per car per day—that is, $R = n \times p$.

Step 4: Since the number of cars rented per day is modeled by the linear function $n(p) = 1000 - 5p$, the revenue R can be represented by the function

$$R(p) = n \times p = (1000 - 5p)p = -5p^2 + 1000p.$$

Step 5: Since the owners plan to charge between \$50 per car per day and \$200 per car per day, the problem is to find the maximum revenue $R(p)$ for p in the closed interval $[50, 200]$.

Step 6: Since R is a continuous function over the closed, bounded interval $[50, 200]$, it has an absolute maximum (and an absolute minimum) in that interval. To find the maximum value, look for critical points. The derivative is $R'(p) = -10p + 1000$. Therefore, the critical point is $p = 100$. When $p = 100$, $R(100) = \$50,000$. When $p = 50$,

$R(p) = \$37,500$. When $p = 200$, $R(p) = \$0$. Therefore, the absolute maximum occurs at $p = \$100$. The car rental company should charge \$100 per day per car to maximize revenue as shown in the following figure.

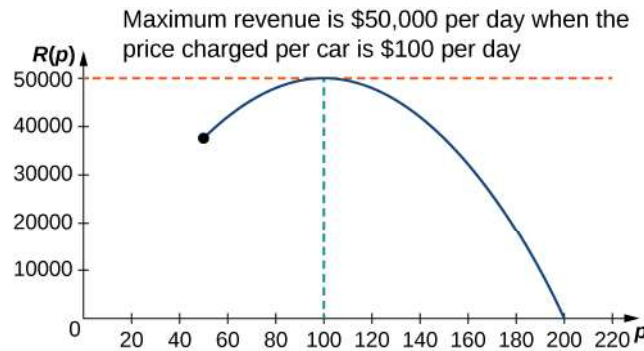


Figure 4.67 To maximize revenue, a car rental company has to balance the price of a rental against the number of cars people will rent at that price.

- ✓ 4.34 A car rental company charges its customers p dollars per day, where $60 \leq p \leq 150$. It has found that the number of cars rented per day can be modeled by the linear function $n(p) = 750 - 5p$. How much should the company charge each customer to maximize revenue?

EXAMPLE 4.36

Maximizing the Area of an Inscribed Rectangle

A rectangle is to be inscribed in the ellipse

$$\frac{x^2}{4} + y^2 = 1.$$

What should the dimensions of the rectangle be to maximize its area? What is the maximum area?

✓ Solution

Step 1: For a rectangle to be inscribed in the ellipse, the sides of the rectangle must be parallel to the axes. Let L be the length of the rectangle and W be its width. Let A be the area of the rectangle.

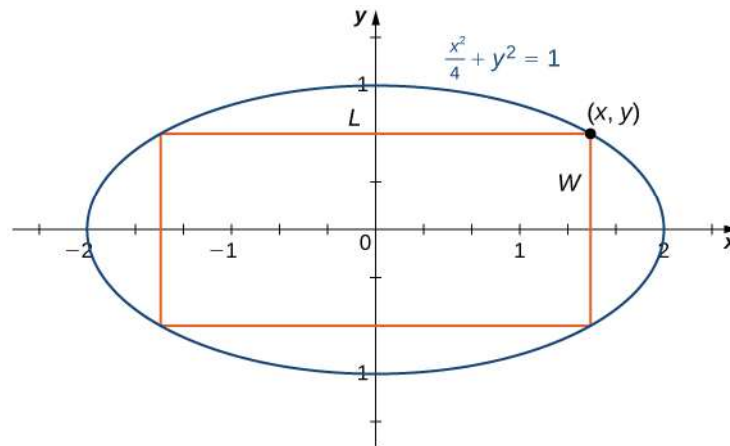


Figure 4.68 We want to maximize the area of a rectangle inscribed in an ellipse.

Step 2: The problem is to maximize A .

Step 3: The area of the rectangle is $A = LW$.

Step 4: Let (x, y) be the corner of the rectangle that lies in the first quadrant, as shown in [Figure 4.68](#). We can write

length $L = 2x$ and width $W = 2y$. Since $\frac{x^2}{4} + y^2 = 1$ and $y > 0$, we have $y = \sqrt{1 - \frac{x^2}{4}}$. Therefore, the area is

$$A = LW = (2x)(2y) = 4x\sqrt{1 - \frac{x^2}{4}} = 2x\sqrt{4 - x^2}.$$

Step 5: From [Figure 4.68](#), we see that to inscribe a rectangle in the ellipse, the x -coordinate of the corner in the first quadrant must satisfy $0 < x < 2$. Therefore, the problem reduces to looking for the maximum value of $A(x)$ over the open interval $(0, 2)$. Since $A(x)$ will have an absolute maximum (and absolute minimum) over the closed interval $[0, 2]$, we consider $A(x) = 2x\sqrt{4 - x^2}$ over the interval $[0, 2]$. If the absolute maximum occurs at an interior point, then we have found an absolute maximum in the open interval.

Step 6: As mentioned earlier, $A(x)$ is a continuous function over the closed, bounded interval $[0, 2]$. Therefore, it has an absolute maximum (and absolute minimum). At the endpoints $x = 0$ and $x = 2$, $A(x) = 0$. For $0 < x < 2$, $A(x) > 0$. Therefore, the maximum must occur at a critical point. Taking the derivative of $A(x)$, we obtain

$$\begin{aligned} A'(x) &= 2\sqrt{4 - x^2} + 2x \cdot \frac{1}{2\sqrt{4 - x^2}}(-2x) \\ &= 2\sqrt{4 - x^2} - \frac{2x^2}{\sqrt{4 - x^2}} \\ &= \frac{8 - 4x^2}{\sqrt{4 - x^2}}. \end{aligned}$$

To find critical points, we need to find where $A'(x) = 0$. We can see that if x is a solution of

$$\frac{8 - 4x^2}{\sqrt{4 - x^2}} = 0, \tag{4.7}$$

then x must satisfy

$$8 - 4x^2 = 0.$$

Therefore, $x^2 = 2$. Thus, $x = \pm\sqrt{2}$ are the possible solutions of [Equation 4.7](#). Since we are considering x over the interval $[0, 2]$, $x = \sqrt{2}$ is a possibility for a critical point, but $x = -\sqrt{2}$ is not. Therefore, we check whether $\sqrt{2}$ is a solution of [Equation 4.7](#). Since $x = \sqrt{2}$ is a solution of [Equation 4.7](#), we conclude that $\sqrt{2}$ is the only critical point of $A(x)$ in the interval $[0, 2]$. Therefore, $A(x)$ must have an absolute maximum at the critical point $x = \sqrt{2}$. To determine the dimensions of the rectangle, we need to find the length L and the width W . If $x = \sqrt{2}$ then

$$y = \sqrt{1 - \frac{(\sqrt{2})^2}{4}} = \sqrt{1 - \frac{1}{2}} = \frac{1}{\sqrt{2}}.$$

Therefore, the dimensions of the rectangle are $L = 2x = 2\sqrt{2}$ and $W = 2y = \frac{2}{\sqrt{2}} = \sqrt{2}$. The area of this rectangle is

$$A = LW = (2\sqrt{2})(\sqrt{2}) = 4.$$



4.35 Modify the area function A if the rectangle is to be inscribed in the unit circle $x^2 + y^2 = 1$. What is the domain of consideration?

Solving Optimization Problems when the Interval Is Not Closed or Is Unbounded

In the previous examples, we considered functions on closed, bounded domains. Consequently, by the extreme value theorem, we were guaranteed that the functions had absolute extrema. Let's now consider functions for which the domain is neither closed nor bounded.

Many functions still have at least one absolute extrema, even if the domain is not closed or the domain is unbounded. For example, the function $f(x) = x^2 + 4$ over $(-\infty, \infty)$ has an absolute minimum of 4 at $x = 0$. Therefore, we can still consider functions over unbounded domains or open intervals and determine whether they have any absolute extrema. In the next example, we try to minimize a function over an unbounded domain. We will see that, although the domain of

consideration is $(0, \infty)$, the function has an absolute minimum.

In the following example, we look at constructing a box of least surface area with a prescribed volume. It is not difficult to show that for a closed-top box, by symmetry, among all boxes with a specified volume, a cube will have the smallest surface area. Consequently, we consider the modified problem of determining which open-topped box with a specified volume has the smallest surface area.

EXAMPLE 4.37

Minimizing Surface Area

A rectangular box with a square base, an open top, and a volume of 216 in.^3 is to be constructed. What should the dimensions of the box be to minimize the surface area of the box? What is the minimum surface area?

Solution

Step 1: Draw a rectangular box and introduce the variable x to represent the length of each side of the square base; let y represent the height of the box. Let S denote the surface area of the open-top box.

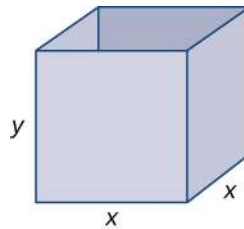


Figure 4.69 We want to minimize the surface area of a square-based box with a given volume.

Step 2: We need to minimize the surface area. Therefore, we need to minimize S .

Step 3: Since the box has an open top, we need only determine the area of the four vertical sides and the base. The area of each of the four vertical sides is $x \cdot y$. The area of the base is x^2 . Therefore, the surface area of the box is

$$S = 4xy + x^2.$$

Step 4: Since the volume of this box is x^2y and the volume is given as 216 in.^3 , the constraint equation is

$$x^2y = 216.$$

Solving the constraint equation for y , we have $y = \frac{216}{x^2}$. Therefore, we can write the surface area as a function of x only:

$$S(x) = 4x \left(\frac{216}{x^2} \right) + x^2.$$

Therefore, $S(x) = \frac{864}{x} + x^2$.

Step 5: Since we are requiring that $x^2y = 216$, we cannot have $x = 0$. Therefore, we need $x > 0$. On the other hand, x is allowed to have any positive value. Note that as x becomes large, the height of the box y becomes correspondingly small so that $x^2y = 216$. Similarly, as x becomes small, the height of the box becomes correspondingly large. We conclude that the domain is the open, unbounded interval $(0, \infty)$. Note that, unlike the previous examples, we cannot reduce our problem to looking for an absolute maximum or absolute minimum over a closed, bounded interval. However, in the next step, we discover why this function must have an absolute minimum over the interval $(0, \infty)$.

Step 6: Note that as $x \rightarrow 0^+$, $S(x) \rightarrow \infty$. Also, as $x \rightarrow \infty$, $S(x) \rightarrow \infty$. Since S is a continuous function that approaches infinity at the ends, it must have an absolute minimum at some $x \in (0, \infty)$. This minimum must occur at a critical point of S . The derivative is

$$S'(x) = -\frac{864}{x^2} + 2x.$$

Therefore, $S'(x) = 0$ when $2x = \frac{864}{x^2}$. Solving this equation for x , we obtain $x^3 = 432$, so $x = \sqrt[3]{432} = 6\sqrt[3]{2}$. Since this

is the only critical point of S , the absolute minimum must occur at $x = 6\sqrt[3]{2}$ (see Figure 4.70). When $x = 6\sqrt[3]{2}$, $y = \frac{216}{(6\sqrt[3]{2})^2} = 3\sqrt[3]{2}$ in. Therefore, the dimensions of the box should be $x = 6\sqrt[3]{2}$ in. and $y = 3\sqrt[3]{2}$ in. With these dimensions, the surface area is

$$S(6\sqrt[3]{2}) = \frac{864}{6\sqrt[3]{2}} + (6\sqrt[3]{2})^2 = 108\sqrt[3]{4} \text{ in.}^2$$

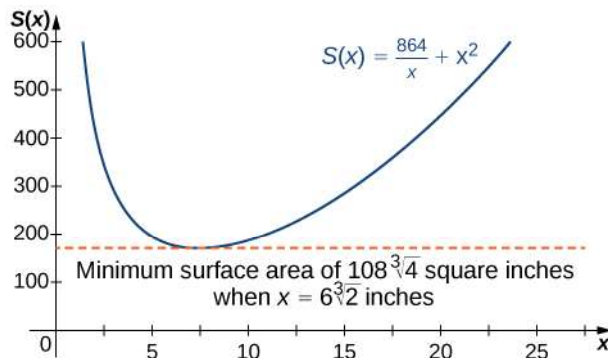


Figure 4.70 We can use a graph to determine the dimensions of a box of given the volume and the minimum surface area.

- ✓ 4.36 Consider the same open-top box, which is to have volume 216 in.^3 . Suppose the cost of the material for the base is 20¢/in.^2 and the cost of the material for the sides is 30¢/in.^2 and we are trying to minimize the cost of this box. Write the cost as a function of the side lengths of the base. (Let x be the side length of the base and y be the height of the box.)



SECTION 4.7 EXERCISES

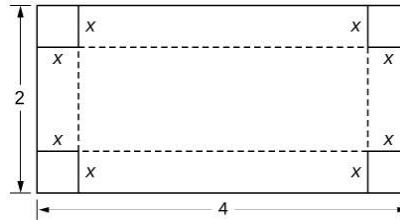
For the following exercises, answer by proof, counterexample, or explanation.

- 311.** When you find the maximum for an optimization problem, why do you need to check the sign of the derivative around the critical points?
- 312.** Why do you need to check the endpoints for optimization problems?
- 313.** *True or False.* For every continuous nonlinear function, you can find the value x that maximizes the function.
- 314.** *True or False.* For every continuous nonconstant function on a closed, finite domain, there exists at least one x that minimizes or maximizes the function.

For the following exercises, set up and evaluate each optimization problem.

- 315.** To carry a suitcase on an airplane, the length + width + height of the box must be less than or equal to 62 in. Assuming the base of the suitcase is square, show that the volume is
- $$V = h\left(31 - \left(\frac{1}{2}\right)h\right)^2.$$
- What height allows you to have the largest volume?

- 316.** You are constructing a cardboard box with the dimensions 2 m by 4 m. You then cut equal-size squares from each corner so you may fold the edges. What are the dimensions of the box with the largest volume?



- 317.** Find the positive integer that minimizes the sum of the number and its reciprocal.

- 318.** Find the two positive integers so that their sum is 10 and the sum of their squares is as large as possible. Then find another two positive integers so that their sum is 10 and the sum of their squares is as small as possible.

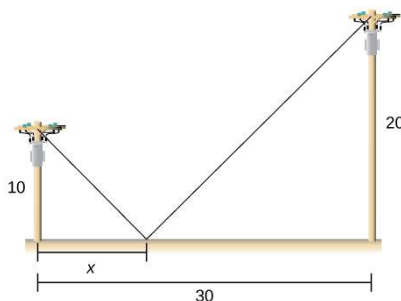
For the following exercises, consider the construction of a pen to enclose an area.

- 319.** You have 400 ft of fencing to construct a rectangular pen for cattle. What are the dimensions of the pen that maximize the area?

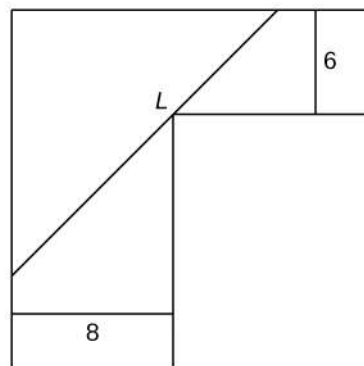
- 320.** You have 800 ft of fencing to make a pen for hogs. If you have a river on one side of your property, what is the dimension of the rectangular pen that maximizes the area?

- 321.** You need to construct a fence around an area of 1600 ft². What are the dimensions of the rectangular pen to minimize the amount of material needed?

- 322.** Two poles are connected by a wire that is also connected to the ground. The first pole is 20 ft tall and the second pole is 10 ft tall. There is a distance of 30 ft between the two poles. Where should the wire be anchored to the ground to minimize the amount of wire needed?

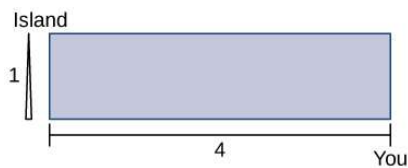


- 323. [T]** You are moving into a new apartment and notice there is a corner where the hallway narrows from 8 ft to 6 ft. What is the length of the longest item that can be carried horizontally around the corner?

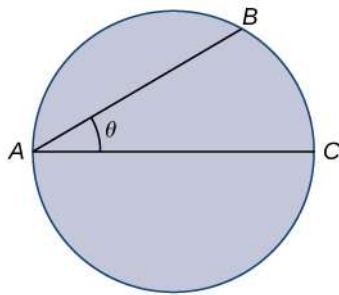


- 324.** A patient's pulse measures 70 bpm, 80 bpm, then 120 bpm. To determine an accurate measurement of pulse, the doctor wants to know what value minimizes the expression $(x - 70)^2 + (x - 80)^2 + (x - 120)^2$? What value minimizes it?

- 326.** You can run at a speed of 6 mph and swim at a speed of 3 mph and are located on the shore, 4 miles east of an island that is 1 mile north of the shoreline. How far should you run west to minimize the time needed to reach the island?



For the following problems, consider a lifeguard at a circular pool with diameter 40 m. He must reach someone who is drowning on the exact opposite side of the pool, at position C. The lifeguard swims with a speed v and runs around the pool at speed $w = 3v$.



- 327.** Find a function that measures the total amount of time it takes to reach the drowning person as a function of the swim angle, θ .
- 328.** Find at what angle θ the lifeguard should swim to reach the drowning person in the least amount of time.
- 329.** A truck uses gas as $g(v) = av + \frac{b}{v}$, where v represents the speed of the truck and g represents the gallons of fuel per mile. Assuming a and b are positive, at what speed is fuel consumption minimized?

For the following exercises, consider a limousine that gets $m(v) = \frac{(120-2v)}{5}$ mi/gal at speed v , the chauffeur costs \$15/h, and gas is \$3.5/gal.

- 330.** Find the cost per mile at speed v .
- 331.** Find the cheapest driving speed.

For the following exercises, consider a pizzeria that sell pizzas for a revenue of $R(x) = ax$ and costs $C(x) = b + cx + dx^2$, where x represents the number of pizzas; $a > c$.

- 332.** Find the profit function for the number of pizzas. How many pizzas gives the largest profit per pizza?
- 333.** Assume that $R(x) = 10x$ and $C(x) = 2x + x^2$. How many pizzas sold maximizes the profit?
- 334.** Assume that $R(x) = 15x$, and $C(x) = 60 + 3x + \frac{1}{2}x^2$. How many pizzas sold maximizes the profit?

For the following exercises, consider a wire 4 ft long cut into two pieces. One piece forms a circle with radius r and the other forms a square of side x .

- 335.** Choose x to maximize the sum of their areas.
- 336.** Choose x to minimize the sum of their areas.

For the following exercises, consider two nonnegative numbers x and y such that $x + y = 10$. Maximize and minimize the quantities.

- 337.** xy
- 338.** x^2y^2
- 339.** $y - \frac{1}{x}$
- 340.** $x^2 - y$

For the following exercises, draw the given optimization problem and solve.

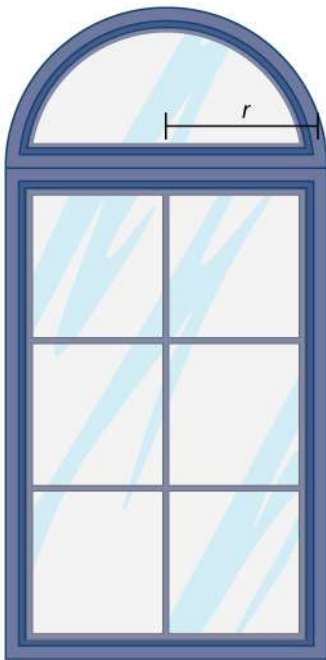
- 341.** Find the volume of the largest right circular cylinder that fits in a sphere of radius 1.
- 342.** Find the volume of the largest right cone that fits in a sphere of radius 1.
- 343.** Find the area of the largest rectangle that fits into the triangle with sides $x = 0$, $y = 0$ and $\frac{x}{4} + \frac{y}{6} = 1$.
- 344.** Find the largest volume of a cylinder that fits into a cone that has base radius R and height h .
- 345.** Find the dimensions of the closed cylinder volume $V = 16\pi$ that has the least amount of surface area.
- 346.** Find the dimensions of a right cone with surface area $S = 4\pi$ that has the largest volume.

For the following exercises, consider the points on the given graphs. Use a calculator to graph the functions.

- 347. [T]** Where is the line $y = 5 - 2x$ closest to the origin?
- 348. [T]** Where is the line $y = 5 - 2x$ closest to point $(1, 1)$?
- 349. [T]** Where is the parabola $y = x^2$ closest to point $(2, 0)$?
- 350. [T]** Where is the parabola $y = x^2$ closest to point $(0, 3)$?

For the following exercises, write an equation that models each related-rates situation. Do not solve.

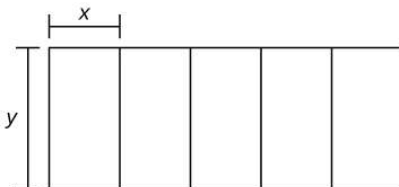
- 351.** A window is composed of a semicircle placed on top of a rectangle. If you have 20 ft of window-framing materials for the outer frame, what is the maximum size of the window you can create? Use r to represent the radius of the semicircle.



- 352.** You have a garden row of 20 watermelon plants that produce an average of 30 watermelons apiece. For any additional watermelon plants planted, the output per watermelon plant drops by one watermelon. How many extra watermelon plants should you plant?

- 353.** You are constructing a box for your cat to sleep in. The plush material for the square bottom of the box costs $\$5/\text{ft}^2$ and the material for the sides costs $\$2/\text{ft}^2$. You need a box with volume 4 ft^3 . Find the dimensions of the box that minimize cost. Use x to represent the length of the side of the box.

- 354.** You are building five identical pens adjacent to each other with a total area of 1000 m^2 , as shown in the following figure. What dimensions should you use to minimize the amount of fencing?



- 355.** You are the manager of an apartment complex with 50 units. When you set rent at $\$800/\text{month}$, all apartments are rented. As you increase rent by $\$25/\text{month}$, one fewer apartment is rented. Maintenance costs run $\$50/\text{month}$ for each occupied unit. What is the rent that maximizes the total amount of profit?

4.8 L'Hôpital's Rule

Learning Objectives

- 4.8.1** Recognize when to apply L'Hôpital's rule.
- 4.8.2** Identify indeterminate forms produced by quotients, products, subtractions, and powers, and apply L'Hôpital's rule in each case.
- 4.8.3** Describe the relative growth rates of functions.

In this section, we examine a powerful tool for evaluating limits. This tool, known as **L'Hôpital's rule**, uses derivatives to calculate limits. With this rule, we will be able to evaluate many limits we have not yet been able to determine. Instead of relying on numerical evidence to conjecture that a limit exists, we will be able to show definitively that a limit exists and to determine its exact value.